

6.6

Work

In everyday life, *work* means an activity that requires muscular or mental effort. In science, the term refers specifically to a force acting on a body and the body's subsequent displacement. This section shows how to calculate work. The applications run from compressing railroad car springs and emptying subterranean tanks to forcing electrons together and lifting satellites into orbit.

Work Done by a Constant Force

When a body moves a distance d along a straight line as a result of being acted on by a force of constant magnitude F in the direction of motion, we define the **work** W done by the force on the body with the formula

$$W = Fd \quad (\text{Constant-force formula for work}). \quad (1)$$

Joules

The joule, abbreviated J and pronounced “jewel,” is named after the English physicist James Prescott Joule (1818–1889). The defining equation is

$$1 \text{ joule} = (1 \text{ newton})(1 \text{ meter}).$$

In symbols, $1 \text{ J} = 1 \text{ N} \cdot \text{m}$.

From Equation (1) we see that the unit of work in any system is the unit of force multiplied by the unit of distance. In SI units (SI stands for *Système International*, or International System), the unit of force is a newton, the unit of distance is a meter, and the unit of work is a newton-meter ($\text{N} \cdot \text{m}$). This combination appears so often it has a special name, the **joule**. In the British system, the unit of work is the foot-pound, a unit frequently used by engineers.

EXAMPLE 1 Jacking Up a Car

If you jack up the side of a 2000-lb car 1.25 ft to change a tire (you have to apply a constant vertical force of about 1000 lb) you will perform $1000 \times 1.25 = 1250$ ft-lb of work on the car. In SI units, you have applied a force of 4448 N through a distance of 0.381 m to do $4448 \times 0.381 \approx 1695$ J of work. ■

Work Done by a Variable Force Along a Line

If the force you apply varies along the way, as it will if you are compressing a spring, the formula $W = Fd$ has to be replaced by an integral formula that takes the variation in F into account.

Suppose that the force performing the work acts along a line that we take to be the x -axis and that its magnitude F is a continuous function of the position. We want to find the work done over the interval from $x = a$ to $x = b$. We partition $[a, b]$ in the usual way and choose an arbitrary point c_k in each subinterval $[x_{k-1}, x_k]$. If the subinterval is short enough, F , being continuous, will not vary much from x_{k-1} to x_k . The amount of work done across the interval will be about $F(c_k)$ times the distance Δx_k , the same as it would be if F were constant and we could apply Equation (1). The total work done from a to b is therefore approximated by the Riemann sum

$$\text{Work} \approx \sum_{k=1}^n F(c_k) \Delta x_k.$$

We expect the approximation to improve as the norm of the partition goes to zero, so we define the work done by the force from a to b to be the integral of F from a to b .

DEFINITION Work

The **work** done by a variable force $F(x)$ directed along the x -axis from $x = a$ to $x = b$ is

$$W = \int_a^b F(x) \, dx. \quad (2)$$

The units of the integral are joules if F is in newtons and x is in meters, and foot-pounds if F is in pounds and x in feet. So, the work done by a force of $F(x) = 1/x^2$ newtons along the x -axis from $x = 1$ m to $x = 10$ m is

$$W = \int_1^{10} \frac{1}{x^2} \, dx = -\frac{1}{x} \Big|_1^{10} = -\frac{1}{10} + 1 = 0.9 \text{ J}.$$

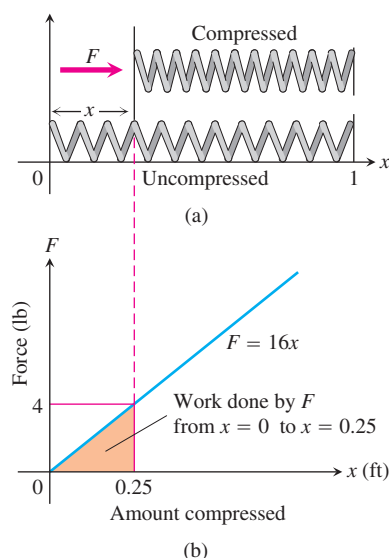


FIGURE 6.58 The force F needed to hold a spring under compression increases linearly as the spring is compressed (Example 2).

Hooke's Law for Springs: $F = kx$

Hooke's Law says that the force it takes to stretch or compress a spring x length units from its natural (unstressed) length is proportional to x . In symbols,

$$F = kx. \quad (3)$$

The constant k , measured in force units per unit length, is a characteristic of the spring, called the **force constant** (or **spring constant**) of the spring. Hooke's Law, Equation (3), gives good results as long as the force doesn't distort the metal in the spring. We assume that the forces in this section are too small to do that.

EXAMPLE 2 Compressing a Spring

Find the work required to compress a spring from its natural length of 1 ft to a length of 0.75 ft if the force constant is $k = 16$ lb/ft.

Solution We picture the uncompressed spring laid out along the x -axis with its movable end at the origin and its fixed end at $x = 1$ ft (Figure 6.58). This enables us to describe the force required to compress the spring from 0 to x with the formula $F = 16x$. To compress the spring from 0 to 0.25 ft, the force must increase from

$$F(0) = 16 \cdot 0 = 0 \text{ lb} \quad \text{to} \quad F(0.25) = 16 \cdot 0.25 = 4 \text{ lb}.$$

The work done by F over this interval is

$$W = \int_0^{0.25} 16x \, dx = 8x^2 \Big|_0^{0.25} = 0.5 \text{ ft}\cdot\text{lb}.$$

Eq. (2) with
 $a = 0, b = 0.25,$
 $F(x) = 16x$

EXAMPLE 3 Stretching a Spring

A spring has a natural length of 1 m. A force of 24 N stretches the spring to a length of 1.8 m.

- Find the force constant k .
- How much work will it take to stretch the spring 2 m beyond its natural length?
- How far will a 45-N force stretch the spring?

Solution

- (a) *The force constant.* We find the force constant from Equation (3). A force of 24 N stretches the spring 0.8 m, so

$$\begin{aligned} 24 &= k(0.8) \\ k &= 24/0.8 = 30 \text{ N/m}. \end{aligned}$$

Eq. (3) with
 $F = 24, x = 0.8$

- (b) *The work to stretch the spring 2 m.* We imagine the unstressed spring hanging along the x -axis with its free end at $x = 0$ (Figure 6.59). The force required to stretch the spring x m beyond its natural length is the force required to pull the free end of the spring x units from the origin. Hooke's Law with $k = 30$ says that this force is

$$F(x) = 30x.$$

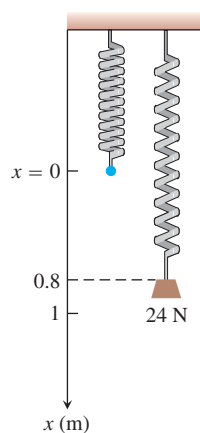


FIGURE 6.59 A 24-N weight stretches this spring 0.8 m beyond its unstressed length (Example 3).

The work done by F on the spring from $x = 0$ m to $x = 2$ m is

$$W = \int_0^2 30x \, dx = 15x^2 \Big|_0^2 = 60 \text{ J.}$$

- (c) *How far will a 45-N force stretch the spring?* We substitute $F = 45$ in the equation $F = 30x$ to find

$$45 = 30x, \quad \text{or} \quad x = 1.5 \text{ m.}$$

A 45-N force will stretch the spring 1.5 m. No calculus is required to find this. ■

The work integral is useful to calculate the work done in lifting objects whose weights vary with their elevation.

EXAMPLE 4 Lifting a Rope and Bucket

A 5-lb bucket is lifted from the ground into the air by pulling in 20 ft of rope at a constant speed (Figure 6.60). The rope weighs 0.08 lb/ft. How much work was spent lifting the bucket and rope?

Solution The bucket has constant weight so the work done lifting it alone is weight $\times$ distance $= 5 \cdot 20 = 100$ ft-lb.

The weight of the rope varies with the bucket's elevation, because less of it is freely hanging. When the bucket is x ft off the ground, the remaining proportion of the rope still being lifted weighs $(0.08) \cdot (20 - x)$ lb. So the work in lifting the rope is

$$\begin{aligned} \text{Work on rope} &= \int_0^{20} (0.08)(20 - x) \, dx = \int_0^{20} (1.6 - 0.08x) \, dx \\ &= \left[1.6x - 0.04x^2 \right]_0^{20} = 32 - 16 = 16 \text{ ft-lb.} \end{aligned}$$

The total work for the bucket and rope combined is

$$100 + 16 = 116 \text{ ft-lb.} \quad \blacksquare$$

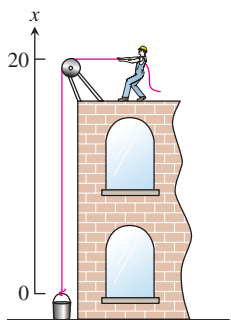


FIGURE 6.60 Lifting the bucket in Example 4.

Pumping Liquids from Containers

How much work does it take to pump all or part of the liquid from a container? To find out, we imagine lifting the liquid out one thin horizontal slab at a time and applying the equation $W = Fd$ to each slab. We then evaluate the integral this leads to as the slabs become thinner and more numerous. The integral we get each time depends on the weight of the liquid and the dimensions of the container, but the way we find the integral is always the same. The next examples show what to do.

EXAMPLE 5 Pumping Oil from a Conical Tank

The conical tank in Figure 6.61 is filled to within 2 ft of the top with olive oil weighing 57 lb/ft³. How much work does it take to pump the oil to the rim of the tank?

Solution We imagine the oil divided into thin slabs by planes perpendicular to the y -axis at the points of a partition of the interval $[0, 8]$.

The typical slab between the planes at y and $y + \Delta y$ has a volume of about

$$\Delta V = \pi(\text{radius})^2(\text{thickness}) = \pi \left(\frac{1}{2}y \right)^2 \Delta y = \frac{\pi}{4} y^2 \Delta y \text{ ft}^3.$$

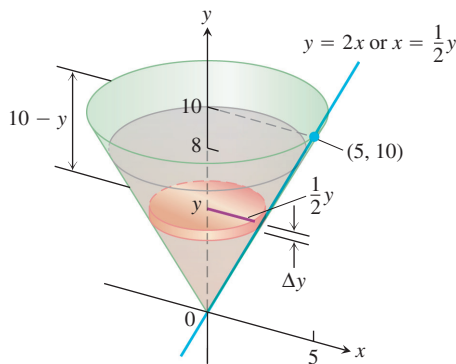


FIGURE 6.61 The olive oil and tank in Example 5.

The force $F(y)$ required to lift this slab is equal to its weight,

$$F(y) = 57 \Delta V = \frac{57\pi}{4} y^2 \Delta y \text{ lb.}$$

Weight = weight per unit
volume $\times$ volume

The distance through which $F(y)$ must act to lift this slab to the level of the rim of the cone is about $(10 - y)$ ft, so the work done lifting the slab is about

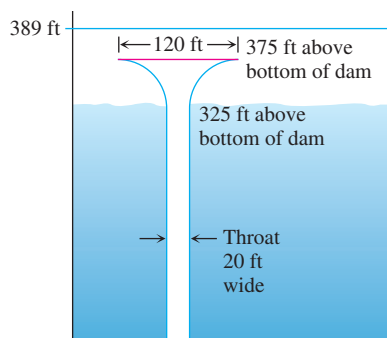
$$\Delta W = \frac{57\pi}{4} (10 - y) y^2 \Delta y \text{ ft-lb.}$$

Assuming there are n slabs associated with the partition of $[0, 8]$, and that $y = y_k$ denotes the plane associated with the k th slab of thickness Δy_k , we can approximate the work done lifting all of the slabs with the Riemann sum

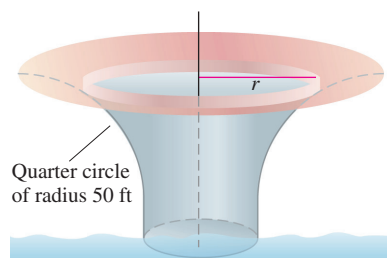
$$W \approx \sum_{k=1}^n \frac{57\pi}{4} (10 - y_k) y_k^2 \Delta y_k \text{ ft-lb.}$$

The work of pumping the oil to the rim is the limit of these sums as the norm of the partition goes to zero.

$$\begin{aligned} W &= \int_0^8 \frac{57\pi}{4} (10 - y) y^2 dy \\ &= \frac{57\pi}{4} \int_0^8 (10y^2 - y^3) dy \\ &= \frac{57\pi}{4} \left[\frac{10y^3}{3} - \frac{y^4}{4} \right]_0^8 \approx 30,561 \text{ ft-lb.} \end{aligned}$$



(a)



(b)

FIGURE 6.62 (a) Cross-section of the glory hole for a dam and (b) the top of the glory hole (Example 6).

EXAMPLE 6 Pumping Water from a Glory Hole

A glory hole is a vertical drain pipe that keeps the water behind a dam from getting too high. The top of the glory hole for a dam is 14 ft below the top of the dam and 375 ft above the bottom (Figure 6.62). The hole needs to be pumped out from time to time to permit the removal of seasonal debris.

From the cross-section in Figure 6.62a, we see that the glory hole is a funnel-shaped drain. The throat of the funnel is 20 ft wide and the head is 120 ft across. The outside boundary of the head cross-section are quarter circles formed with 50-ft radii, shown in Figure 6.62b. The glory hole is formed by rotating a cross-section around its center. Consequently, all horizontal cross-sections are circular disks throughout the entire glory hole. We calculate the work required to pump water from

- the throat of the hole.
- the funnel portion.

Solution

- Pumping from the throat.* A typical slab in the throat between the planes at y and $y + \Delta y$ has a volume of about

$$\Delta V = \pi(\text{radius})^2(\text{thickness}) = \pi(10)^2 \Delta y \text{ ft}^3.$$

The force $F(y)$ required to lift this slab is equal to its weight (about 62.4 lb/ft³ for water),

$$F(y) = 62.4 \Delta V = 6240\pi \Delta y \text{ lb.}$$

The distance through which $F(y)$ must act to lift this slab to the top of the hole is $(375 - y)$ ft, so the work done lifting the slab is

$$\Delta W = 6240\pi(375 - y)\Delta y \text{ ft-lb.}$$

We can approximate the work done in pumping the water from the throat by summing the work done lifting all the slabs individually, and then taking the limit of this Riemann sum as the norm of the partition goes to zero. This gives the integral

$$\begin{aligned} W &= \int_0^{325} 6240\pi(375 - y) dy \\ &= 6240\pi \left[375y - \frac{y^2}{2} \right]_0^{325} \\ &\approx 1,353,869,354 \text{ ft-lb.} \end{aligned}$$

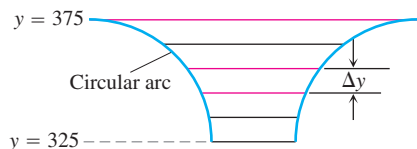


FIGURE 6.63 The glory hole funnel portion.

(b) Pumping from the funnel. To compute the work necessary to pump water from the funnel portion of the glory hole, from $y = 325$ to $y = 375$, we need to compute ΔV for approximating elements in the funnel as shown in Figure 6.63. As can be seen from the figure, the radii of the slabs vary with height y .

In Exercises 33 and 34, you are asked to complete the analysis to determine the total work required to pump the water and to find the power of the pumps necessary to pump out the glory hole. ■

EXERCISES 6.6

Springs

- 1. Spring constant** It took 1800 J of work to stretch a spring from its natural length of 2 m to a length of 5 m. Find the spring's force constant.
- 2. Stretching a spring** A spring has a natural length of 10 in. An 800-lb force stretches the spring to 14 in.
 - a.** Find the force constant.
 - b.** How much work is done in stretching the spring from 10 in. to 12 in.?
 - c.** How far beyond its natural length will a 1600-lb force stretch the spring?
- 3. Stretching a rubber band** A force of 2 N will stretch a rubber band 2 cm (0.02 m). Assuming that Hooke's Law applies, how far will a 4-N force stretch the rubber band? How much work does it take to stretch the rubber band this far?
- 4. Stretching a spring** If a force of 90 N stretches a spring 1 m beyond its natural length, how much work does it take to stretch the spring 5 m beyond its natural length?
- 5. Subway car springs** It takes a force of 21,714 lb to compress a coil spring assembly on a New York City Transit Authority subway car from its free height of 8 in. to its fully compressed height of 5 in.

- a.** What is the assembly's force constant?
- b.** How much work does it take to compress the assembly the first half inch? the second half inch? Answer to the nearest in.-lb.

(Data courtesy of Bombardier, Inc., Mass Transit Division, for spring assemblies in subway cars delivered to the New York City Transit Authority from 1985 to 1987.)

- 6. Bathroom scale** A bathroom scale is compressed $1/16$ in. when a 150-lb person stands on it. Assuming that the scale behaves like a spring that obeys Hooke's Law, how much does someone who compresses the scale $1/8$ in. weigh? How much work is done compressing the scale $1/8$ in.?

Work Done By a Variable Force

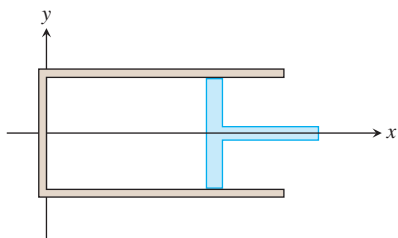
- 7. Lifting a rope** A mountain climber is about to haul up a 50 m length of hanging rope. How much work will it take if the rope weighs 0.624 N/m?
- 8. Leaky sandbag** A bag of sand originally weighing 144 lb was lifted at a constant rate. As it rose, sand also leaked out at a constant rate. The sand was half gone by the time the bag had been

lifted to 18 ft. How much work was done lifting the sand this far? (Neglect the weight of the bag and lifting equipment.)

- 9. Lifting an elevator cable** An electric elevator with a motor at the top has a multistrand cable weighing 4.5 lb/ft. When the car is at the first floor, 180 ft of cable are paid out, and effectively 0 ft are out when the car is at the top floor. How much work does the motor do just lifting the cable when it takes the car from the first floor to the top?
- 10. Force of attraction** When a particle of mass m is at $(x, 0)$, it is attracted toward the origin with a force whose magnitude is k/x^2 . If the particle starts from rest at $x = b$ and is acted on by no other forces, find the work done on it by the time it reaches $x = a$, $0 < a < b$.
- 11. Compressing gas** Suppose that the gas in a circular cylinder of cross-sectional area A is being compressed by a piston. If p is the pressure of the gas in pounds per square inch and V is the volume in cubic inches, show that the work done in compressing the gas from state (p_1, V_1) to state (p_2, V_2) is given by the equation

$$\text{Work} = \int_{(p_1, V_1)}^{(p_2, V_2)} p \, dV.$$

(Hint: In the coordinates suggested in the figure here, $dV = A \, dx$. The force against the piston is pA .)



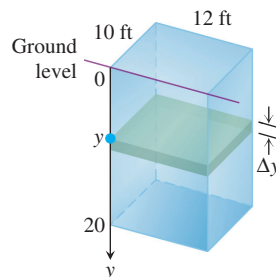
- 12. (Continuation of Exercise 11.)** Use the integral in Exercise 11 to find the work done in compressing the gas from $V_1 = 243 \text{ in.}^3$ to $V_2 = 32 \text{ in.}^3$ if $p_1 = 50 \text{ lb/in.}^2$ and p and V obey the gas law $pV^{1.4} = \text{constant}$ (for adiabatic processes).
- 13. Leaky bucket** Assume the bucket in Example 4 is leaking. It starts with 2 gal of water (16 lb) and leaks at a constant rate. It finishes draining just as it reaches the top. How much work was spent lifting the water alone? (Hint: Do not include the rope and bucket, and find the proportion of water left at elevation x ft.)
- 14. (Continuation of Exercise 13.)** The workers in Example 4 and Exercise 13 changed to a larger bucket that held 5 gal (40 lb) of water, but the new bucket had an even larger leak so that it, too, was empty by the time it reached the top. Assuming that the water leaked out at a steady rate, how much work was done lifting the water alone? (Do not include the rope and bucket.)

Pumping Liquids from Containers

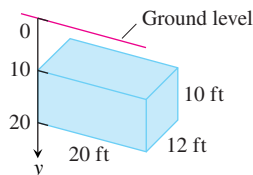
The Weight of Water

Because of Earth's rotation and variations in its gravitational field, the weight of a cubic foot of water at sea level can vary from about 62.26 lb at the equator to as much as 62.59 lb near the poles, a variation of about 0.5%. A cubic foot that weighs about 62.4 lb in Melbourne and New York City will weigh 62.5 lb in Juneau and Stockholm. Although 62.4 is a typical figure and common textbook value, there is considerable variation.

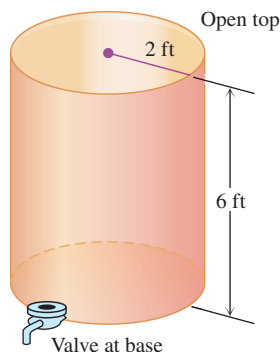
- 15. Pumping water** The rectangular tank shown here, with its top at ground level, is used to catch runoff water. Assume that the water weighs 62.4 lb/ft³.
- How much work does it take to empty the tank by pumping the water back to ground level once the tank is full?
 - If the water is pumped to ground level with a (5/11)-horsepower (hp) motor (work output 250 ft-lb/sec), how long will it take to empty the full tank (to the nearest minute)?
 - Show that the pump in part (b) will lower the water level 10 ft (halfway) during the first 25 min of pumping.
 - The weight of water** What are the answers to parts (a) and (b) in a location where water weighs 62.26 lb/ft³? 62.59 lb/ft³?



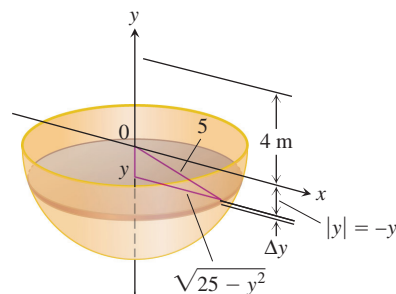
- 16. Emptying a cistern** The rectangular cistern (storage tank for rainwater) shown below has its top 10 ft below ground level. The cistern, currently full, is to be emptied for inspection by pumping its contents to ground level.
- How much work will it take to empty the cistern?
 - How long will it take a 1/2 hp pump, rated at 275 ft-lb/sec, to pump the tank dry?
 - How long will it take the pump in part (b) to empty the tank halfway? (It will be less than half the time required to empty the tank completely.)
 - The weight of water** What are the answers to parts (a) through (c) in a location where water weighs 62.26 lb/ft³? 62.59 lb/ft³?



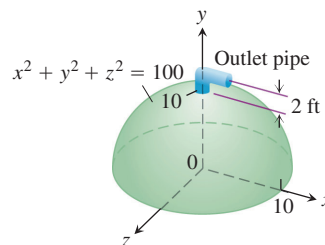
- 17. Pumping oil** How much work would it take to pump oil from the tank in Example 5 to the level of the top of the tank if the tank were completely full?
- 18. Pumping a half-full tank** Suppose that, instead of being full, the tank in Example 5 is only half full. How much work does it take to pump the remaining oil to a level 4 ft above the top of the tank?
- 19. Emptying a tank** A vertical right circular cylindrical tank measures 30 ft high and 20 ft in diameter. It is full of kerosene weighing 51.2 lb/ft^3 . How much work does it take to pump the kerosene to the level of the top of the tank?
- 20.** The cylindrical tank shown here can be filled by pumping water from a lake 15 ft below the bottom of the tank. There are two ways to go about it. One is to pump the water through a hose attached to a valve in the bottom of the tank. The other is to attach the hose to the rim of the tank and let the water pour in. Which way will be faster? Give reasons for your answer.



- 21. a. Pumping milk** Suppose that the conical container in Example 5 contains milk (weighing 64.5 lb/ft^3) instead of olive oil. How much work will it take to pump the contents to the rim?
- b. Pumping oil** How much work will it take to pump the oil in Example 5 to a level 3 ft above the cone's rim?
- 22. Pumping seawater** To design the interior surface of a huge stainless-steel tank, you revolve the curve $y = x^2$, $0 \leq x \leq 4$, about the y -axis. The container, with dimensions in meters, is to be filled with seawater, which weighs $10,000 \text{ N/m}^3$. How much work will it take to empty the tank by pumping the water to the tank's top?
- 23. Emptying a water reservoir** We model pumping from spherical containers the way we do from other containers, with the axis of integration along the vertical axis of the sphere. Use the figure here to find how much work it takes to empty a full hemispherical water reservoir of radius 5 m by pumping the water to a height of 4 m above the top of the reservoir. Water weighs 9800 N/m^3 .



- 24.** You are in charge of the evacuation and repair of the storage tank shown here. The tank is a hemisphere of radius 10 ft and is full of benzene weighing 56 lb/ft^3 . A firm you contacted says it can empty the tank for $1/2\epsilon$ per foot-pound of work. Find the work required to empty the tank by pumping the benzene to an outlet 2 ft above the top of the tank. If you have \$5000 budgeted for the job, can you afford to hire the firm?



Work and Kinetic Energy

- 25. Kinetic energy** If a variable force of magnitude $F(x)$ moves a body of mass m along the x -axis from x_1 to x_2 , the body's velocity v can be written as dx/dt (where t represents time). Use Newton's second law of motion $F = m(dv/dt)$ and the Chain Rule

$$\frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$$

to show that the net work done by the force in moving the body from x_1 to x_2 is

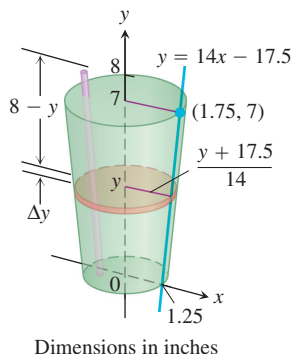
$$W = \int_{x_1}^{x_2} F(x) dx = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2,$$

where v_1 and v_2 are the body's velocities at x_1 and x_2 . In physics, the expression $(1/2)mv^2$ is called the *kinetic energy* of a body of mass m moving with velocity v . Therefore, *the work done by the force equals the change in the body's kinetic energy*, and we can find the work by calculating this change.

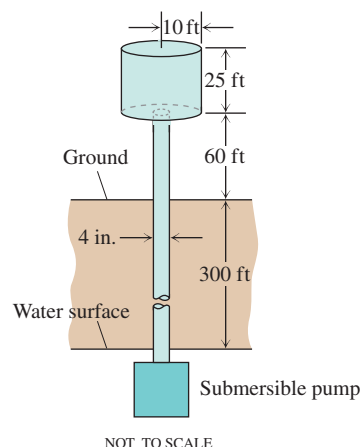
In Exercises 26–32, use the result of Exercise 25.

- 26. Tennis** A 2-oz tennis ball was served at 160 ft/sec (about 109 mph). How much work was done on the ball to make it go this fast? (To find the ball's mass from its weight, express the weight in pounds and divide by 32 ft/sec^2 , the acceleration of gravity.)
- 27. Baseball** How many foot-pounds of work does it take to throw a baseball 90 mph? A baseball weighs 5 oz, or 0.3125 lb .

- 28. Golf** A 1.6-oz golf ball is driven off the tee at a speed of 280 ft/sec (about 191 mph). How many foot-pounds of work are done on the ball getting it into the air?
- 29. Tennis** During the match in which Pete Sampras won the 1990 U.S. Open men's tennis championship, Sampras hit a serve that was clocked at a phenomenal 124 mph. How much work did Sampras have to do on the 2-oz ball to get it to that speed?
- 30. Football** A quarterback threw a 14.5-oz football 88 ft/sec (60 mph). How many foot-pounds of work were done on the ball to get it to this speed?
- 31. Softball** How much work has to be performed on a 6.5-oz softball to pitch it 132 ft/sec (90 mph)?
- 32. A ball bearing** A 2-oz steel ball bearing is placed on a vertical spring whose force constant is $k = 18$ lb/ft. The spring is compressed 2 in. and released. About how high does the ball bearing go?
- 33. Pumping the funnel of the glory hole** (Continuation of Example 6.)
- Find the radius of the cross-section (funnel portion) of the glory hole in Example 6 as a function of the height y above the floor of the dam (from $y = 325$ to $y = 375$).
 - Find ΔV for the funnel section of the glory hole (from $y = 325$ to $y = 375$).
 - Find the work necessary to pump out the funnel section by formulating and evaluating the appropriate definite integral.
- 34. Pumping water from a glory hole** (Continuation of Exercise 33.)
- Find the total work necessary to pump out the glory hole, by adding the work necessary to pump both the throat and funnel sections.
 - Your answer to part (a) is in foot-pounds. A more useful form is horsepower-hours, since motors are rated in horsepower. To convert from foot-pounds to horsepower-hours, divide by 1.98×10^6 . How many hours would it take a 1000-horsepower motor to pump out the glory hole, assuming that the motor was fully efficient?
- 35. Drinking a milkshake** The truncated conical container shown here is full of strawberry milkshake that weighs $4/9$ oz/in.³ As you can see, the container is 7 in. deep, 2.5 in. across at the base, and 3.5 in. across at the top (a standard size at Brigham's in Boston). The straw sticks up an inch above the top. About how much work does it take to suck up the milkshake through the straw (neglecting friction)? Answer in inch-ounces.



- 36. Water tower** Your town has decided to drill a well to increase its water supply. As the town engineer, you have determined that a water tower will be necessary to provide the pressure needed for distribution, and you have designed the system shown here. The water is to be pumped from a 300 ft well through a vertical 4 in. pipe into the base of a cylindrical tank 20 ft in diameter and 25 ft high. The base of the tank will be 60 ft aboveground. The pump is a 3 hp pump, rated at 1650 ft · lb/sec. To the nearest hour, how long will it take to fill the tank the first time? (Include the time it takes to fill the pipe.) Assume that water weighs 62.4 lb/ft³.



- 37. Putting a satellite in orbit** The strength of Earth's gravitational field varies with the distance r from Earth's center, and the magnitude of the gravitational force experienced by a satellite of mass m during and after launch is

$$F(r) = \frac{mMG}{r^2}.$$

Here, $M = 5.975 \times 10^{24}$ kg is Earth's mass, $G = 6.6720 \times 10^{-11}$ N · m² kg⁻² is the universal gravitational constant, and r is measured in meters. The work it takes to lift a 1000-kg satellite from Earth's surface to a circular orbit 35,780 km above Earth's center is therefore given by the integral

$$\text{Work} = \int_{6,370,000}^{35,780,000} \frac{1000MG}{r^2} dr \text{ joules.}$$

Evaluate the integral. The lower limit of integration is Earth's radius in meters at the launch site. (This calculation does not take into account energy spent lifting the launch vehicle or energy spent bringing the satellite to orbit velocity.)

- 38. Forcing electrons together** Two electrons r meters apart repel each other with a force of

$$F = \frac{23 \times 10^{-29}}{r^2} \text{ newtons.}$$

- Suppose one electron is held fixed at the point $(1, 0)$ on the x -axis (units in meters). How much work does it take to move a second electron along the x -axis from the point $(-1, 0)$ to the origin?
- Suppose an electron is held fixed at each of the points $(-1, 0)$ and $(1, 0)$. How much work does it take to move a third electron along the x -axis from $(5, 0)$ to $(3, 0)$?

6.7 Fluid Pressures and Forces

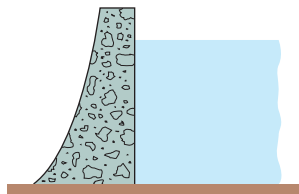


FIGURE 6.64 To withstand the increasing pressure, dams are built thicker as they go down.

We make dams thicker at the bottom than at the top (Figure 6.64) because the pressure against them increases with depth. The pressure at any point on a dam depends only on how far below the surface the point is and not on how much the surface of the dam happens to be tilted at that point. The pressure, in pounds per square foot at a point h feet below the surface, is always $62.4h$. The number 62.4 is the weight-density of water in pounds per cubic foot. The pressure h feet below the surface of any fluid is the fluid's *weight-density* times h .

The Pressure-Depth Equation

In a fluid that is standing still, the pressure p at depth h is the fluid's weight-density w times h :

$$p = wh. \quad (1)$$

In this section we use the equation $p = wh$ to derive a formula for the total force exerted by a fluid against all or part of a vertical or horizontal containing wall.

Weight-density

A fluid's weight-density is its weight per unit volume. Typical values (lb/ft^3) are

Gasoline	42
Mercury	849
Milk	64.5
Molasses	100
Olive oil	57
Seawater	64
Water	62.4

The Constant-Depth Formula for Fluid Force

In a container of fluid with a flat horizontal base, the total force exerted by the fluid against the base can be calculated by multiplying the area of the base by the pressure at the base. We can do this because total force equals force per unit area (pressure) times area. (See Figure 6.65.) If F , p , and A are the total force, pressure, and area, then

$$\begin{aligned} F &= \text{total force} = \text{force per unit area} \times \text{area} \\ &= \text{pressure} \times \text{area} = pA \\ &= whA. \end{aligned}$$

$p = wh$ from
Eq. (1)

Fluid Force on a Constant-Depth Surface

$$F = pA = whA \quad (2)$$

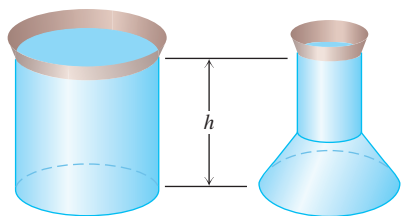


FIGURE 6.65 These containers are filled with water to the same depth and have the same base area. The total force is therefore the same on the bottom of each container. The containers' shapes do not matter here.

For example, the weight-density of water is $62.4 \text{ lb}/\text{ft}^3$, so the fluid force at the bottom of a $10 \text{ ft} \times 20 \text{ ft}$ rectangular swimming pool 3 ft deep is

$$\begin{aligned} F &= whA = (62.4 \text{ lb}/\text{ft}^3)(3 \text{ ft})(10 \cdot 20 \text{ ft}^2) \\ &= 37,440 \text{ lb}. \end{aligned}$$

For a flat plate submerged *horizontally*, like the bottom of the swimming pool just discussed, the downward force acting on its upper face due to liquid pressure is given by Equation (2). If the plate is submerged *vertically*, however, then the pressure against it will be different at different depths and Equation (2) no longer is usable in that form (because h varies). By dividing the plate into many narrow horizontal bands or strips, we can create a Riemann sum whose limit is the fluid force against the side of the submerged vertical plate. Here is the procedure.

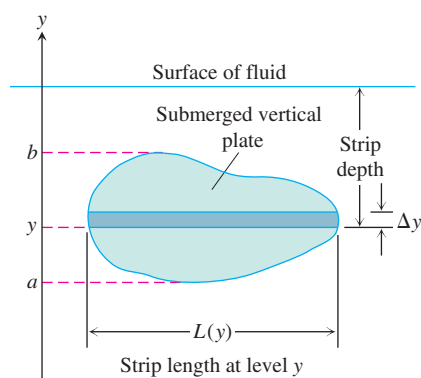


FIGURE 6.66 The force exerted by a fluid against one side of a thin, flat horizontal strip is about $\Delta F = \text{pressure} \times \text{area} = w \times (\text{strip depth}) \times L(y) \Delta y$.

The Variable-Depth Formula

Suppose we want to know the force exerted by a fluid against one side of a vertical plate submerged in a fluid of weight-density w . To find it, we model the plate as a region extending from $y = a$ to $y = b$ in the xy -plane (Figure 6.66). We partition $[a, b]$ in the usual way and imagine the region to be cut into thin horizontal strips by planes perpendicular to the y -axis at the partition points. The typical strip from y to $y + \Delta y$ is Δy units wide by $L(y)$ units long. We assume $L(y)$ to be a continuous function of y .

The pressure varies across the strip from top to bottom. If the strip is narrow enough, however, the pressure will remain close to its bottom-edge value of $w \times (\text{strip depth})$. The force exerted by the fluid against one side of the strip will be about

$$\Delta F = (\text{pressure along bottom edge}) \times (\text{area})$$

$$= w \cdot (\text{strip depth}) \cdot L(y) \Delta y.$$

Assume there are n strips associated with the partition of $a \leq y \leq b$ and that y_k is the bottom edge of the k th strip having length $L(y_k)$ and width Δy_k . The force against the entire plate is approximated by summing the forces against each strip, giving the Riemann sum

$$F \approx \sum_{k=1}^n (w \cdot (\text{strip depth})_k \cdot L(y_k)) \Delta y_k. \quad (3)$$

The sum in Equation (3) is a Riemann sum for a continuous function on $[a, b]$, and we expect the approximations to improve as the norm of the partition goes to zero. The force against the plate is the limit of these sums.

The Integral for Fluid Force Against a Vertical Flat Plate

Suppose that a plate submerged vertically in fluid of weight-density w runs from $y = a$ to $y = b$ on the y -axis. Let $L(y)$ be the length of the horizontal strip measured from left to right along the surface of the plate at level y . Then the force exerted by the fluid against one side of the plate is

$$F = \int_a^b w \cdot (\text{strip depth}) \cdot L(y) dy. \quad (4)$$

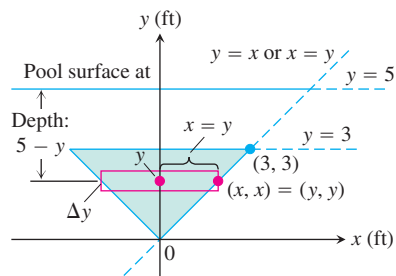


FIGURE 6.67 To find the force on one side of the submerged plate in Example 1, we can use a coordinate system like the one here.

EXAMPLE 1 Applying the Integral for Fluid Force

A flat isosceles right triangular plate with base 6 ft and height 3 ft is submerged vertically, base up, 2 ft below the surface of a swimming pool. Find the force exerted by the water against one side of the plate.

Solution We establish a coordinate system to work in by placing the origin at the plate's bottom vertex and running the y -axis upward along the plate's axis of symmetry (Figure 6.67). The surface of the pool lies along the line $y = 5$ and the plate's top edge along the line $y = 3$. The plate's right-hand edge lies along the line $y = x$, with the upper right vertex at $(3, 3)$. The length of a thin strip at level y is

$$L(y) = 2x = 2y.$$

The depth of the strip beneath the surface is $(5 - y)$. The force exerted by the water against one side of the plate is therefore

$$\begin{aligned}
 F &= \int_a^b w \cdot \left(\frac{\text{strip}}{\text{depth}} \right) \cdot L(y) \, dy && \text{Eq. (4)} \\
 &= \int_0^3 62.4(5 - y)2y \, dy \\
 &= 124.8 \int_0^3 (5y - y^2) \, dy \\
 &= 124.8 \left[\frac{5}{2}y^2 - \frac{y^3}{3} \right]_0^3 = 1684.8 \text{ lb.}
 \end{aligned}$$

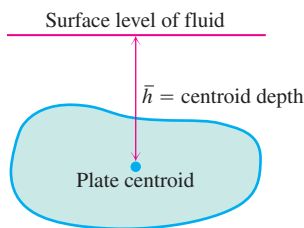


FIGURE 6.68 The force against one side of the plate is $w \cdot \bar{h} \cdot \text{plate area}$.

Fluid Forces and Centroids

If we know the location of the centroid of a submerged flat vertical plate (Figure 6.68), we can take a shortcut to find the force against one side of the plate. From Equation (4),

$$\begin{aligned}
 F &= \int_a^b w \times (\text{strip depth}) \times L(y) \, dy \\
 &= w \int_a^b (\text{strip depth}) \times L(y) \, dy \\
 &= w \times (\text{moment about surface level line of region occupied by plate}) \\
 &= w \times (\text{depth of plate's centroid}) \times (\text{area of plate}).
 \end{aligned}$$

Fluid Forces and Centroids

The force of a fluid of weight-density w against one side of a submerged flat vertical plate is the product of w , the distance $\bar{h}$ from the plate's centroid to the fluid surface, and the plate's area:

$$F = w\bar{h}A. \quad (5)$$

EXAMPLE 2 Finding Fluid Force Using Equation (5)

Use Equation (5) to find the force in Example 1.

Solution The centroid of the triangle (Figure 6.67) lies on the y -axis, one-third of the way from the base to the vertex, so $\bar{h} = 3$. The triangle's area is

$$\begin{aligned}
 A &= \frac{1}{2}(\text{base})(\text{height}) \\
 &= \frac{1}{2}(6)(3) = 9.
 \end{aligned}$$

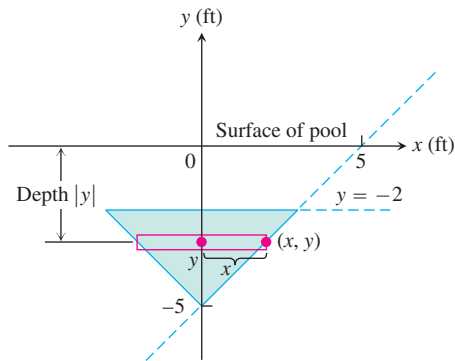
Hence,

$$\begin{aligned}
 F &= w\bar{h}A = (62.4)(3)(9) \\
 &= 1684.8 \text{ lb.}
 \end{aligned}$$

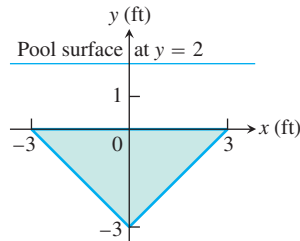
EXERCISES 6.7

The weight-densities of the fluids in the following exercises can be found in the table on page 456.

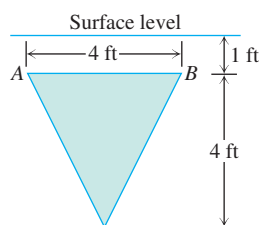
1. **Triangular plate** Calculate the fluid force on one side of the plate in Example 1 using the coordinate system shown here.



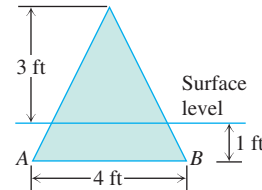
2. **Triangular plate** Calculate the fluid force on one side of the plate in Example 1 using the coordinate system shown here.



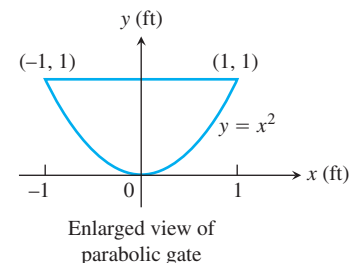
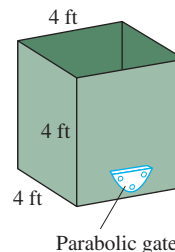
3. **Lowered triangular plate** The plate in Example 1 is lowered another 2 ft into the water. What is the fluid force on one side of the plate now?
4. **Raised triangular plate** The plate in Example 1 is raised to put its top edge at the surface of the pool. What is the fluid force on one side of the plate now?
5. **Triangular plate** The isosceles triangular plate shown here is submerged vertically 1 ft below the surface of a freshwater lake.
- Find the fluid force against one face of the plate.
 - What would be the fluid force on one side of the plate if the water were seawater instead of freshwater?



6. **Rotated triangular plate** The plate in Exercise 5 is revolved 180° about line AB so that part of the plate sticks out of the lake, as shown here. What force does the water exert on one face of the plate now?

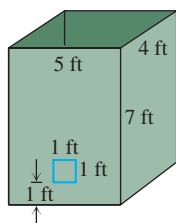


7. **New England Aquarium** The viewing portion of the rectangular glass window in a typical fish tank at the New England Aquarium in Boston is 63 in. wide and runs from 0.5 in. below the water's surface to 33.5 in. below the surface. Find the fluid force against this portion of the window. The weight-density of seawater is 64 lb/ft^3 . (In case you were wondering, the glass is $3/4$ in. thick and the tank walls extend 4 in. above the water to keep the fish from jumping out.)
8. **Fish tank** A horizontal rectangular freshwater fish tank with base $2 \text{ ft} \times 4 \text{ ft}$ and height 2 ft (interior dimensions) is filled to within 2 in. of the top.
- Find the fluid force against each side and end of the tank.
 - If the tank is sealed and stood on end (without spilling), so that one of the square ends is the base, what does that do to the fluid forces on the rectangular sides?
9. **Semicircular plate** A semicircular plate 2 ft in diameter sticks straight down into freshwater with the diameter along the surface. Find the force exerted by the water on one side of the plate.
10. **Milk truck** A tank truck hauls milk in a 6-ft-diameter horizontal right circular cylindrical tank. How much force does the milk exert on each end of the tank when the tank is half full?
11. The cubical metal tank shown here has a parabolic gate, held in place by bolts and designed to withstand a fluid force of 160 lb without rupturing. The liquid you plan to store has a weight-density of 50 lb/ft^3 .
- What is the fluid force on the gate when the liquid is 2 ft deep?
 - What is the maximum height to which the container can be filled without exceeding its design limitation?

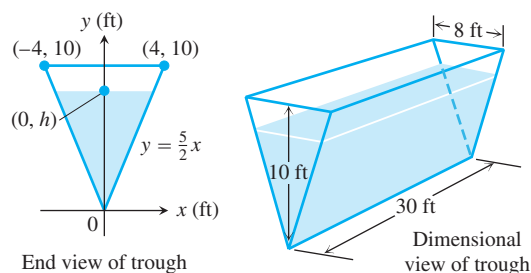


12. The rectangular tank shown here has a $1 \text{ ft} \times 1 \text{ ft}$ square window 1 ft above the base. The window is designed to withstand a fluid force of 312 lb without cracking.

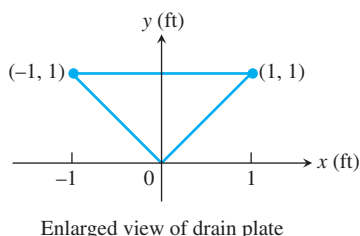
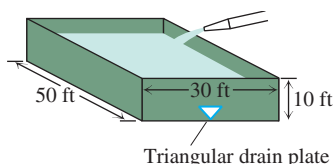
- What fluid force will the window have to withstand if the tank is filled with water to a depth of 3 ft?
- To what level can the tank be filled with water without exceeding the window's design limitation?



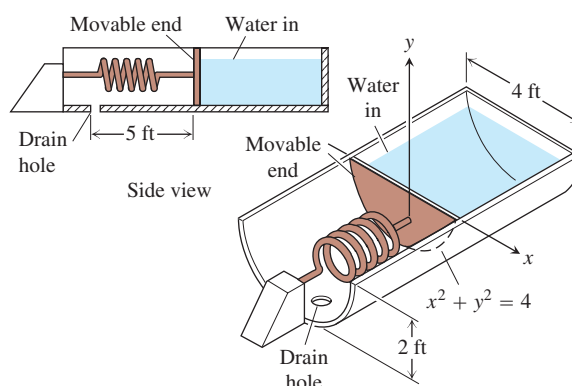
13. The end plates of the trough shown here were designed to withstand a fluid force of 6667 lb. How many cubic feet of water can the tank hold without exceeding this limitation? Round down to the nearest cubic foot.



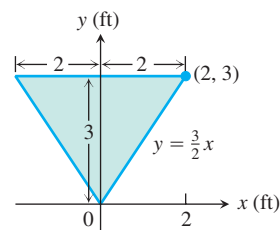
14. Water is running into the rectangular swimming pool shown here at the rate of $1000 \text{ ft}^3/\text{h}$.
- Find the fluid force against the triangular drain plate after 9 h of filling.
 - The drain plate is designed to withstand a fluid force of 520 lb. How high can you fill the pool without exceeding this limitation?



15. A vertical rectangular plate a units long by b units wide is submerged in a fluid of weight-density w with its long edges parallel to the fluid's surface. Find the average value of the pressure along the vertical dimension of the plate. Explain your answer.
16. (Continuation of Exercise 15.) Show that the force exerted by the fluid on one side of the plate is the average value of the pressure (found in Exercise 15) times the area of the plate.
17. Water pours into the tank here at the rate of $4 \text{ ft}^3/\text{min}$. The tank's cross-sections are 4-ft-diameter semicircles. One end of the tank is movable, but moving it to increase the volume compresses a spring. The spring constant is $k = 100 \text{ lb}/\text{ft}$. If the end of the tank moves 5 ft against the spring, the water will drain out of a safety hole in the bottom at the rate of $5 \text{ ft}^3/\text{min}$. Will the movable end reach the hole before the tank overflows?



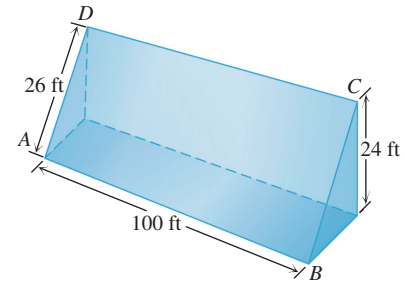
18. **Watering trough** The vertical ends of a watering trough are squares 3 ft on a side.
- Find the fluid force against the ends when the trough is full.
 - How many inches do you have to lower the water level in the trough to reduce the fluid force by 25%?
19. **Milk carton** A rectangular milk carton measures 3.75 in. $\times$ 3.75 in. at the base and is 7.75 in. tall. Find the force of the milk on one side when the carton is full.
20. **Olive oil can** A standard olive oil can measures 5.75 in. $\times$ 3.5 in. at the base and is 10 in. tall. Find the fluid force against the base and each side when the can is full.
21. **Watering trough** The vertical ends of a watering trough are isosceles triangles like the one shown here (dimensions in feet).



- Find the fluid force against the ends when the trough is full.

- b. How many inches do you have to lower the water level in the trough to cut the fluid force on the ends in half? (Answer to the nearest half-inch.)
- c. Does it matter how long the trough is? Give reasons for your answer.
22. The face of a dam is a rectangle, $ABCD$, of dimensions $AB = CD = 100$ ft, $AD = BC = 26$ ft. Instead of being vertical, the plane $ABCD$ is inclined as indicated in the accompanying figure, so that the top of the dam is 24 ft higher than the bottom.

Find the force due to water pressure on the dam when the surface of the water is level with the top of the dam.



Chapter 6

Additional and Advanced Exercises

Volume and Length

1. A solid is generated by revolving about the x -axis the region bounded by the graph of the positive continuous function $y = f(x)$, the x -axis, and the fixed line $x = a$ and the variable line $x = b$, $b > a$. Its volume, for all b , is $b^2 - ab$. Find $f(x)$.
2. A solid is generated by revolving about the x -axis the region bounded by the graph of the positive continuous function $y = f(x)$, the x -axis, and the lines $x = 0$ and $x = a$. Its volume, for all $a > 0$, is $a^2 + a$. Find $f(x)$.
3. Suppose that the increasing function $f(x)$ is smooth for $x \geq 0$ and that $f(0) = a$. Let $s(x)$ denote the length of the graph of f from $(0, a)$ to $(x, f(x))$, $x > 0$. Find $f(x)$ if $s(x) = Cx$ for some constant C . What are the allowable values for C ?
4. a. Show that for $0 < \alpha \leq \pi/2$,

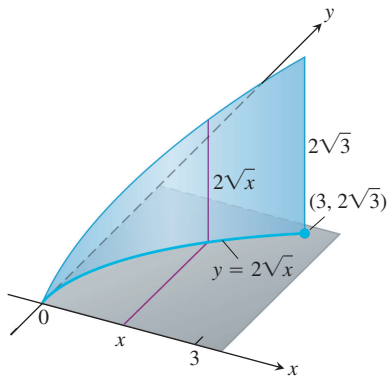
$$\int_0^\alpha \sqrt{1 + \cos^2 \theta} \, d\theta > \sqrt{\alpha^2 + \sin^2 \alpha}.$$

- b. Generalize the result in part (a).

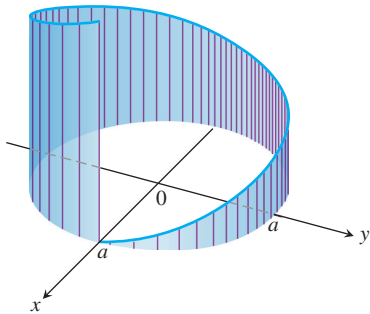
Moments and Centers of Mass

5. Find the centroid of the region bounded below by the x -axis and above by the curve $y = 1 - x^n$, n an even positive integer. What is the limiting position of the centroid as $n \rightarrow \infty$?
6. If you haul a telephone pole on a two-wheeled carriage behind a truck, you want the wheels to be 3 ft or so behind the pole's center of mass to provide an adequate "tongue" weight. NYNEX's class 1.40-ft wooden poles have a 27-in. circumference at the top and a 43.5-in. circumference at the base. About how far from the top is the center of mass?
7. Suppose that a thin metal plate of area A and constant density δ occupies a region R in the xy -plane, and let M_y be the plate's moment about the y -axis. Show that the plate's moment about the line $x = b$ is
 - a. $M_y - b\delta A$ if the plate lies to the right of the line, and
 - b. $b\delta A - M_y$ if the plate lies to the left of the line.
8. Find the center of mass of a thin plate covering the region bounded by the curve $y^2 = 4ax$ and the line $x = a$, a positive constant, if the density at (x, y) is directly proportional to (a) x , (b) $|y|$.
9. a. Find the centroid of the region in the first quadrant bounded by two concentric circles and the coordinate axes, if the circles have radii a and b , $0 < a < b$, and their centers are at the origin.
b. Find the limits of the coordinates of the centroid as a approaches b and discuss the meaning of the result.
10. A triangular corner is cut from a square 1 ft on a side. The area of the triangle removed is 36 in.^2 . If the centroid of the remaining region is 7 in. from one side of the original square, how far is it from the remaining sides?
11. At points on the curve $y = 2\sqrt{x}$, line segments of length $h = y$ are drawn perpendicular to the xy -plane. (See accompanying figure.) Find the area of the surface formed by these perpendiculars from $(0, 0)$ to $(3, 2\sqrt{3})$.

Surface Area



12. At points on a circle of radius a , line segments are drawn perpendicular to the plane of the circle, the perpendicular at each point P being of length ks , where s is the length of the arc of the circle measured counterclockwise from $(a, 0)$ to P and k is a positive constant, as shown here. Find the area of the surface formed by the perpendiculars along the arc beginning at $(a, 0)$ and extending once around the circle.



Work

13. A particle of mass m starts from rest at time $t = 0$ and is moved along the x -axis with constant acceleration a from $x = 0$ to $x = h$ against a variable force of magnitude $F(t) = t^2$. Find the work done.
14. **Work and kinetic energy** Suppose a 1.6-oz golf ball is placed on a vertical spring with force constant $k = 2$ lb/in. The spring is compressed 6 in. and released. About how high does the ball go (measured from the spring's rest position)?

Fluid Force

15. A triangular plate ABC is submerged in water with its plane vertical. The side AB , 4 ft long, is 6 ft below the surface of the water, while the vertex C is 2 ft below the surface. Find the force exerted by the water on one side of the plate.
16. A vertical rectangular plate is submerged in a fluid with its top edge parallel to the fluid's surface. Show that the force exerted by the fluid on one side of the plate equals the average value of the pressure up and down the plate times the area of the plate.
17. The *center of pressure* on one side of a plane region submerged in a fluid is defined to be the point at which the total force exerted by the fluid can be applied without changing its total moment about any axis in the plane. Find the depth to the center of pressure (a) on a vertical rectangle of height h and width b if its upper edge is in the surface of the fluid; (b) on a vertical triangle of height h and base b if the vertex opposite b is a ft and the base b is $(a + h)$ ft below the surface of the fluid.

Chapter 6

Practice Exercises

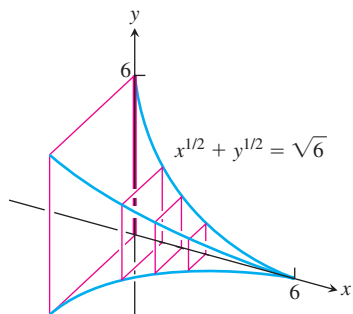
Volumes

Find the volumes of the solids in Exercises 1–16.

1. The solid lies between planes perpendicular to the x -axis at $x = 0$ and $x = 1$. The cross-sections perpendicular to the x -axis between these planes are circular disks whose diameters run from the parabola $y = x^2$ to the parabola $y = \sqrt{x}$.
2. The base of the solid is the region in the first quadrant between the line $y = x$ and the parabola $y = 2\sqrt{x}$. The cross-sections of

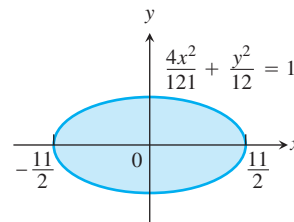
the solid perpendicular to the x -axis are equilateral triangles whose bases stretch from the line to the curve.

3. The solid lies between planes perpendicular to the x -axis at $x = \pi/4$ and $x = 5\pi/4$. The cross-sections between these planes are circular disks whose diameters run from the curve $y = 2 \cos x$ to the curve $y = 2 \sin x$.
4. The solid lies between planes perpendicular to the x -axis at $x = 0$ and $x = 6$. The cross-sections between these planes are squares whose bases run from the x -axis up to the curve $x^{1/2} + y^{1/2} = \sqrt{6}$.



5. The solid lies between planes perpendicular to the x -axis at $x = 0$ and $x = 4$. The cross-sections of the solid perpendicular to the x -axis between these planes are circular disks whose diameters run from the curve $x^2 = 4y$ to the curve $y^2 = 4x$.
6. The base of the solid is the region bounded by the parabola $y^2 = 4x$ and the line $x = 1$ in the xy -plane. Each cross-section perpendicular to the x -axis is an equilateral triangle with one edge in the plane. (The triangles all lie on the same side of the plane.)
7. Find the volume of the solid generated by revolving the region bounded by the x -axis, the curve $y = 3x^4$, and the lines $x = 1$ and $x = -1$ about (a) the x -axis; (b) the y -axis; (c) the line $x = 1$; (d) the line $y = 3$.
8. Find the volume of the solid generated by revolving the “triangular” region bounded by the curve $y = 4/x^3$ and the lines $x = 1$ and $y = 1/2$ about (a) the x -axis; (b) the y -axis; (c) the line $x = 2$; (d) the line $y = 4$.
9. Find the volume of the solid generated by revolving the region bounded on the left by the parabola $x = y^2 + 1$ and on the right by the line $x = 5$ about (a) the x -axis; (b) the y -axis; (c) the line $x = 5$.
10. Find the volume of the solid generated by revolving the region bounded by the parabola $y^2 = 4x$ and the line $y = x$ about (a) the x -axis; (b) the y -axis; (c) the line $x = 4$; (d) the line $y = 4$.
11. Find the volume of the solid generated by revolving the “triangular” region bounded by the x -axis, the line $x = \pi/3$, and the curve $y = \tan x$ in the first quadrant about the x -axis.
12. Find the volume of the solid generated by revolving the region bounded by the curve $y = \sin x$ and the lines $x = 0$, $x = \pi$, and $y = 2$ about the line $y = 2$.
13. Find the volume of the solid generated by revolving the region between the x -axis and the curve $y = x^2 - 2x$ about (a) the x -axis; (b) the line $y = -1$; (c) the line $x = 2$; (d) the line $y = 2$.

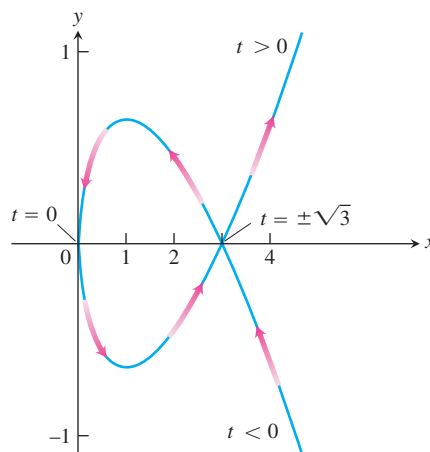
14. Find the volume of the solid generated by revolving about the x -axis the region bounded by $y = 2 \tan x$, $y = 0$, $x = -\pi/4$, and $x = \pi/4$. (The region lies in the first and third quadrants and resembles a skewed bowtie.)
15. **Volume of a solid sphere hole** A round hole of radius $\sqrt{3}$ ft is bored through the center of a solid sphere of a radius 2 ft. Find the volume of material removed from the sphere.
16. **Volume of a football** The profile of a football resembles the ellipse shown here. Find the football’s volume to the nearest cubic inch.



Lengths of Curves

Find the lengths of the curves in Exercises 17–23.

17. $y = x^{1/2} - (1/3)x^{3/2}$, $1 \leq x \leq 4$
18. $x = y^{2/3}$, $1 \leq y \leq 8$
19. $y = (5/12)x^{6/5} - (5/8)x^{4/5}$, $1 \leq x \leq 32$
20. $x = (y^3/12) + (1/y)$, $1 \leq y \leq 2$
21. $x = 5 \cos t - \cos 5t$, $y = 5 \sin t - \sin 5t$, $0 \leq t \leq \pi/2$
22. $x = t^3 - 6t^2$, $y = t^3 + 6t^2$, $0 \leq t \leq 1$
23. $x = 3 \cos \theta$, $y = 3 \sin \theta$, $0 \leq \theta \leq \frac{3\pi}{2}$
24. Find the length of the enclosed loop $x = t^2$, $y = (t^3/3) - t$ shown here. The loop starts at $t = -\sqrt{3}$ and ends at $t = \sqrt{3}$.



Centroids and Centers of Mass

25. Find the centroid of a thin, flat plate covering the region enclosed by the parabolas $y = 2x^2$ and $y = 3 - x^2$.

26. Find the centroid of a thin, flat plate covering the region enclosed by the x -axis, the lines $x = 2$ and $x = -2$, and the parabola $y = x^2$.
27. Find the centroid of a thin, flat plate covering the “triangular” region in the first quadrant bounded by the y -axis, the parabola $y = x^2/4$, and the line $y = 4$.
28. Find the centroid of a thin, flat plate covering the region enclosed by the parabola $y^2 = x$ and the line $x = 2y$.
29. Find the center of mass of a thin, flat plate covering the region enclosed by the parabola $y^2 = x$ and the line $x = 2y$ if the density function is $\delta(y) = 1 + y$. (Use horizontal strips.)
30. a. Find the center of mass of a thin plate of constant density covering the region between the curve $y = 3/x^{3/2}$ and the x -axis from $x = 1$ to $x = 9$.
- b. Find the plate's center of mass if, instead of being constant, the density is $\delta(x) = x$. (Use vertical strips.)

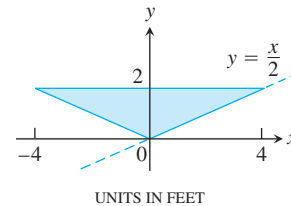
Areas of Surfaces of Revolution

In Exercises 31–36, find the areas of the surfaces generated by revolving the curves about the given axes.

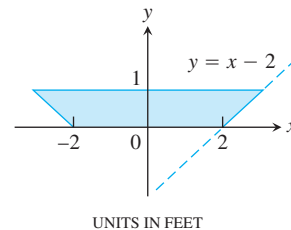
31. $y = \sqrt{2x + 1}$, $0 \leq x \leq 3$; x -axis
32. $y = x^3/3$, $0 \leq x \leq 1$; x -axis
33. $x = \sqrt{4y - y^2}$, $1 \leq y \leq 2$; y -axis
34. $x = \sqrt{y}$, $2 \leq y \leq 6$; y -axis
35. $x = t^2/2$, $y = 2t$, $0 \leq t \leq \sqrt{5}$; x -axis
36. $x = t^2 + 1/(2t)$, $y = 4\sqrt{t}$, $1/\sqrt{2} \leq t \leq 1$; y -axis

Work

37. **Lifting equipment** A rock climber is about to haul up 100 N (about 22.5 lb) of equipment that has been hanging beneath her on 40 m of rope that weighs 0.8 newton per meter. How much work will it take? (*Hint:* Solve for the rope and equipment separately, then add.)
38. **Leaky tank truck** You drove an 800-gal tank truck of water from the base of Mt. Washington to the summit and discovered on arrival that the tank was only half full. You started with a full tank, climbed at a steady rate, and accomplished the 4750-ft elevation change in 50 min. Assuming that the water leaked out at a steady rate, how much work was spent in carrying water to the top? Do not count the work done in getting yourself and the truck there. Water weighs 8 lb/U.S. gal.
39. **Stretching a spring** If a force of 20 lb is required to hold a spring 1 ft beyond its unstressed length, how much work does it take to stretch the spring this far? An additional foot?
40. **Garage door spring** A force of 200 N will stretch a garage door spring 0.8 m beyond its unstressed length. How far will a 300-N force stretch the spring? How much work does it take to stretch the spring this far from its unstressed length?
41. **Pumping a reservoir** A reservoir shaped like a right circular cone, point down, 20 ft across the top and 8 ft deep, is full of water. How much work does it take to pump the water to a level 6 ft above the top?
42. **Pumping a reservoir** (*Continuation of Exercise 41.*) The reservoir is filled to a depth of 5 ft, and the water is to be pumped to the same level as the top. How much work does it take?
43. **Pumping a conical tank** A right circular conical tank, point down, with top radius 5 ft and height 10 ft is filled with a liquid whose weight-density is 60 lb/ft³. How much work does it take to pump the liquid to a point 2 ft above the tank? If the pump is driven by a motor rated at 275 ft-lb/sec (1/2 hp), how long will it take to empty the tank?
44. **Pumping a cylindrical tank** A storage tank is a right circular cylinder 20 ft long and 8 ft in diameter with its axis horizontal. If the tank is half full of olive oil weighing 57 lb/ft³, find the work done in emptying it through a pipe that runs from the bottom of the tank to an outlet that is 6 ft above the top of the tank.
45. **Trough of water** The vertical triangular plate shown here is the end plate of a trough full of water ($w = 62.4$). What is the fluid force against the plate?

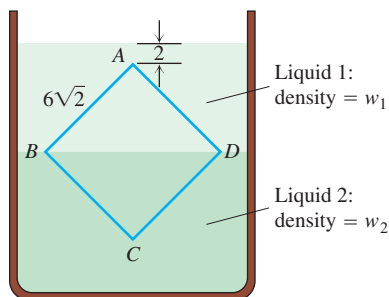


46. **Trough of maple syrup** The vertical trapezoid plate shown here is the end plate of a trough full of maple syrup weighing 75 lb/ft³. What is the force exerted by the syrup against the end plate of the trough when the syrup is 10 in. deep?

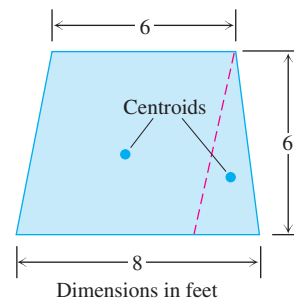


47. **Force on a parabolic gate** A flat vertical gate in the face of a dam is shaped like the parabolic region between the curve $y = 4x^2$ and the line $y = 4$, with measurements in feet. The top of the gate lies 5 ft below the surface of the water. Find the force exerted by the water against the gate ($w = 62.4$).
- T** 48. You plan to store mercury ($w = 849$ lb/ft³) in a vertical rectangular tank with a 1 ft square base side whose interior side wall can withstand a total fluid force of 40,000 lb. About how many cubic feet of mercury can you store in the tank at any one time?

49. The container profiled in the accompanying figure is filled with two nonmixing liquids of weight-density w_1 and w_2 . Find the fluid force on one side of the vertical square plate $ABCD$. The points B and D lie in the boundary layer and the square is $6\sqrt{2}$ ft on a side.



50. The isosceles trapezoidal plate shown here is submerged vertically in water ($w = 62.4$) with its upper edge 4 ft below the surface. Find the fluid force on one side of the plate in two different ways:



- By evaluating an integral.
- By dividing the plate into a parallelogram and an isosceles triangle, locating their centroids, and using the equation $F = \bar{w}\bar{h}A$ from Section 6.7.

Chapter 6**Questions to Guide Your Review**

1. How do you define and calculate the volumes of solids by the method of slicing? Give an example.
2. How are the disk and washer methods for calculating volumes derived from the method of slicing? Give examples of volume calculations by these methods.
3. Describe the method of cylindrical shells. Give an example.
4. How do you define the length of a smooth parametrized curve $x = f(t)$, $y = g(t)$, $a \leq t \leq b$? What does smoothness have to do with length? What else do you need to know about the parametrization to find the curve's length? Give examples.
5. How do you find the length of the graph of a smooth function over a closed interval? Give an example. What about functions that do not have continuous first derivatives?
6. What is a center of mass?
7. How do you locate the center of mass of a straight, narrow rod or strip of material? Give an example. If the density of the material is constant, you can tell right away where the center of mass is. Where is it?
8. How do you locate the center of mass of a thin flat plate of material? Give an example.
9. How do you define and calculate the area of the surface swept out by revolving the graph of a smooth function $y = f(x)$, $a \leq x \leq b$, about the x -axis? Give an example.
10. Under what conditions can you find the area of the surface generated by revolving a curve $x = f(t)$, $y = g(t)$, $a \leq t \leq b$, about the x -axis? The y -axis? Give examples.
11. What do Pappus's two theorems say? Give examples of how they are used to calculate surface areas and volumes and to locate centroids.
12. How do you define and calculate the work done by a variable force directed along a portion of the x -axis? How do you calculate the work it takes to pump a liquid from a tank? Give examples.
13. How do you calculate the force exerted by a liquid against a portion of a vertical wall? Give an example.

Chapter 6

Technology Application Projects

Mathematica/Maple Module

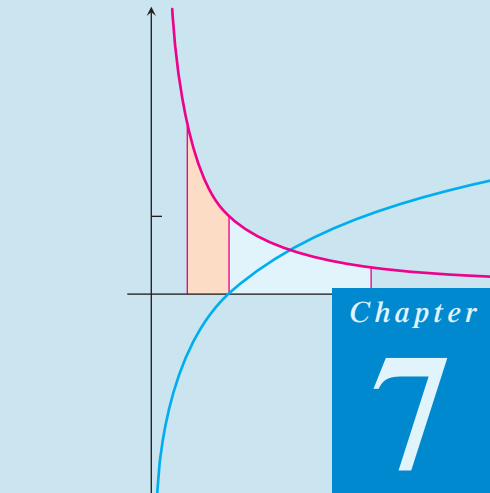
Using Riemann Sums to Estimate Areas, Volumes, and Lengths of Curves

Visualize and approximate areas and volumes in **Part I** and **Part II**: Volumes of Revolution; and **Part III**: Lengths of Curves.

Mathematica/Maple Module

Modeling a Bungee Cord Jump

Collect data (or use data previously collected) to build and refine a model for the force exerted by a jumper's bungee cord. Use the work-energy theorem to compute the distance fallen for a given jumper and a given length of bungee cord.



Chapter 7

TRANSCENDENTAL FUNCTIONS

OVERVIEW Functions can be classified into two broad groups (see Section 1.4). Polynomial functions are called *algebraic*, as are functions obtained from them by addition, multiplication, division, or taking powers and roots. Functions that are not algebraic are called *transcendental*. The trigonometric, exponential, logarithmic, and hyperbolic functions are transcendental, as are their inverses.

Transcendental functions occur frequently in many calculus settings and applications, including growths of populations, vibrations and waves, efficiencies of computer algorithms, and the stability of engineered structures. In this chapter we introduce several important transcendental functions and investigate their graphs, properties, derivatives, and integrals.

7.1

Inverse Functions and Their Derivatives

A function that undoes, or inverts, the effect of a function f is called the *inverse* of f . Many common functions, though not all, are paired with an inverse. Important inverse functions often show up in formulas for antiderivatives and solutions of differential equations. Inverse functions also play a key role in the development and properties of the logarithmic and exponential functions, as we will see in Section 7.3.

One-to-One Functions

A function is a rule that assigns a value from its range to each element in its domain. Some functions assign the same range value to more than one element in the domain. The function $f(x) = x^2$ assigns the same value, 1, to both of the numbers -1 and $+1$; the sines of $\pi/3$ and $2\pi/3$ are both $\sqrt{3}/2$. Other functions assume each value in their range no more than once. The square roots and cubes of different numbers are always different. A function that has distinct values at distinct elements in its domain is called one-to-one. These functions take on any one value in their range exactly once.

DEFINITION One-to-One Function

A function $f(x)$ is **one-to-one** on a domain D if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ in D .

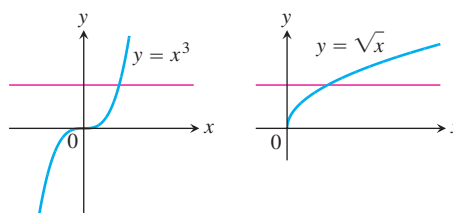
EXAMPLE 1 Domains of One-to-One Functions

- (a) $f(x) = \sqrt{x}$ is one-to-one on any domain of nonnegative numbers because $\sqrt{x_1} \neq \sqrt{x_2}$ whenever $x_1 \neq x_2$.
- (b) $g(x) = \sin x$ is *not* one-to-one on the interval $[0, \pi]$ because $\sin(\pi/6) = \sin(5\pi/6)$. The sine *is* one-to-one on $[0, \pi/2]$, however, because it is a strictly increasing function on $[0, \pi/2]$. ■

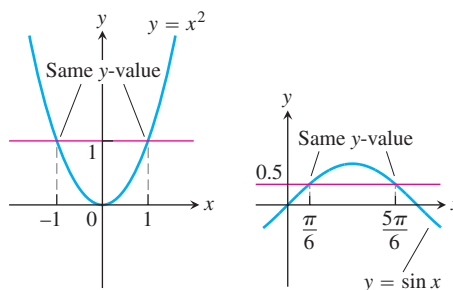
The graph of a one-to-one function $y = f(x)$ can intersect a given horizontal line at most once. If it intersects the line more than once, it assumes the same y -value more than once, and is therefore not one-to-one (Figure 7.1).

The Horizontal Line Test for One-to-One Functions

A function $y = f(x)$ is one-to-one if and only if its graph intersects each horizontal line at most once.



One-to-one: Graph meets each horizontal line at most once.



Not one-to-one: Graph meets one or more horizontal lines more than once.

FIGURE 7.1 Using the horizontal line test, we see that $y = x^3$ and $y = \sqrt{x}$ are one-to-one on their domains $(-\infty, \infty)$ and $[0, \infty)$, but $y = x^2$ and $y = \sin x$ are not one-to-one on their domains $(-\infty, \infty)$.

Inverse Functions

Since each output of a one-to-one function comes from just one input, the effect of the function can be inverted to send an output back to the input from which it came.

DEFINITION Inverse Function

Suppose that f is a one-to-one function on a domain D with range R . The **inverse function** f^{-1} is defined by

$$f^{-1}(a) = b \text{ if } f(b) = a.$$

The domain of f^{-1} is R and the range of f^{-1} is D .

The domains and ranges of f and f^{-1} are interchanged. The symbol f^{-1} for the inverse of f is read “ f inverse.” The “ -1 ” in f^{-1} is *not* an exponent: $f^{-1}(x)$ does not mean $1/f(x)$.

If we apply f to send an input x to the output $f(x)$ and follow by applying f^{-1} to $f(x)$ we get right back to x , just where we started. Similarly, if we take some number y in the range of f , apply f^{-1} to it, and then apply f to the resulting value $f^{-1}(y)$, we get back the value y with which we began. Composing a function and its inverse has the same effect as doing nothing.

$$(f^{-1} \circ f)(x) = x, \quad \text{for all } x \text{ in the domain of } f$$

$$(f \circ f^{-1})(y) = y, \quad \text{for all } y \text{ in the domain of } f^{-1} \text{ (or range of } f)$$

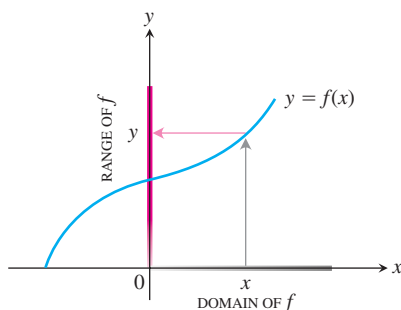
Only a one-to-one function can have an inverse. The reason is that if $f(x_1) = y$ and $f(x_2) = y$ for two distinct inputs x_1 and x_2 , then there is no way to assign a value to $f^{-1}(y)$ that satisfies both $f^{-1}(f(x_1)) = x_1$ and $f^{-1}(f(x_2)) = x_2$.

A function that is increasing on an interval, satisfying $f(x_2) > f(x_1)$ when $x_2 > x_1$, is one-to-one and has an inverse. Decreasing functions also have an inverse (Exercise 39). Functions that have positive derivatives at all x are increasing (Corollary 3 of the Mean Value Theorem, Section 4.2), and so they have inverses. Similarly, functions with negative derivatives at all x are decreasing and have inverses. Functions that are neither increasing nor decreasing may still be one-to-one and have an inverse, as with the function $\sec^{-1} x$ in Section 7.7.

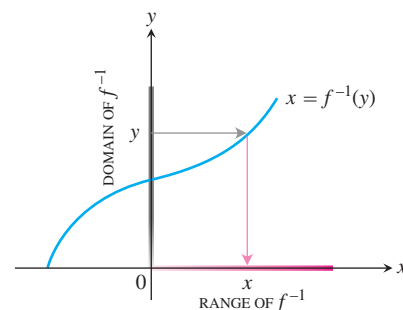
Finding Inverses

The graphs of a function and its inverse are closely related. To read the value of a function from its graph, we start at a point x on the x -axis, go vertically to the graph, and then move horizontally to the y -axis to read the value of y . The inverse function can be read from the graph by reversing this process. Start with a point y on the y -axis, go horizontally to the graph, and then move vertically to the x -axis to read the value of $x = f^{-1}(y)$ (Figure 7.2).

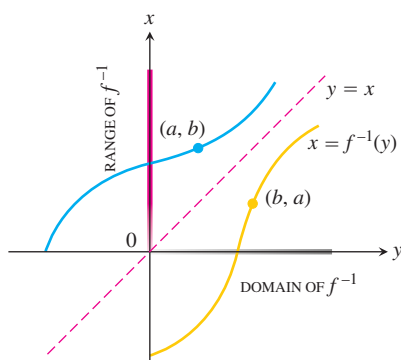
We want to set up the graph of f^{-1} so that its input values lie along the x -axis, as is usually done for functions, rather than on the y -axis. To achieve this we interchange the x and y axes by reflecting across the 45° line $y = x$. After this reflection we have a new graph that represents f^{-1} . The value of $f^{-1}(x)$ can now be read from the graph in the usual way, by starting with a point x on the x -axis, going vertically to the graph and then horizontally to the y -axis to get the value of $f^{-1}(x)$. Figure 7.2 indicates the relation between the graphs of f and f^{-1} . The graphs are interchanged by reflection through the line $y = x$.



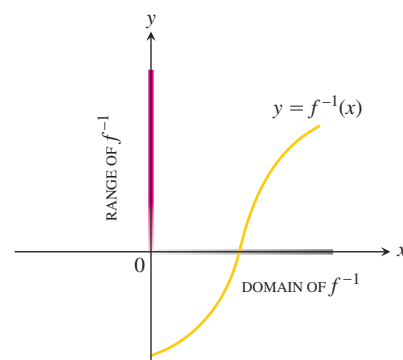
(a) To find the value of f at x , we start at x , go up to the curve, and then over to the y -axis.



(b) The graph of f is already the graph of f^{-1} , but with x and y interchanged. To find the x that gave y , we start at y and go over to the curve and down to the x -axis. The domain of f^{-1} is the range of f . The range of f^{-1} is the domain of f .



(c) To draw the graph of f^{-1} in the more usual way, we reflect the system in the line $y = x$.



(d) Then we interchange the letters x and y . We now have a normal-looking graph of f^{-1} as a function of x .

FIGURE 7.2 Determining the graph of $y = f^{-1}(x)$ from the graph of $y = f(x)$.

The process of passing from f to f^{-1} can be summarized as a two-step process.

1. Solve the equation $y = f(x)$ for x . This gives a formula $x = f^{-1}(y)$ where x is expressed as a function of y .
2. Interchange x and y , obtaining a formula $y = f^{-1}(x)$ where f^{-1} is expressed in the conventional format with x as the independent variable and y as the dependent variable.

EXAMPLE 2 Finding an Inverse Function

Find the inverse of $y = \frac{1}{2}x + 1$, expressed as a function of x .

Solution

1. Solve for x in terms of y :

$$y = \frac{1}{2}x + 1$$

$$2y = x + 2$$

$$x = 2y - 2.$$

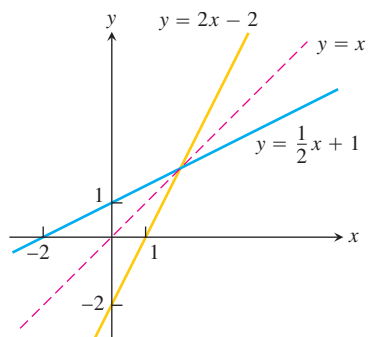


FIGURE 7.3 Graphing $f(x) = (1/2)x + 1$ and $f^{-1}(x) = 2x - 2$ together shows the graphs' symmetry with respect to the line $y = x$. The slopes are reciprocals of each other (Example 2).

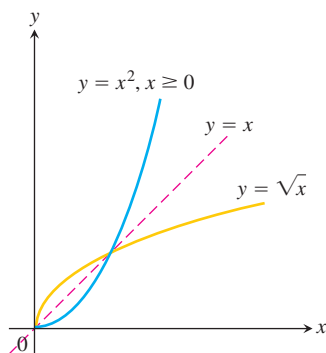


FIGURE 7.4 The functions $y = \sqrt{x}$ and $y = x^2, x \geq 0$, are inverses of one another (Example 3).

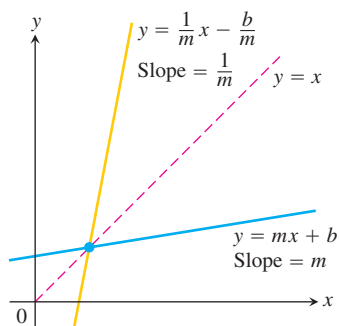


FIGURE 7.5 The slopes of nonvertical lines reflected through the line $y = x$ are reciprocals of each other.

2. *Interchange x and y :* $y = 2x - 2$.

The inverse of the function $f(x) = (1/2)x + 1$ is the function $f^{-1}(x) = 2x - 2$. To check, we verify that both composites give the identity function:

$$f^{-1}(f(x)) = 2\left(\frac{1}{2}x + 1\right) - 2 = x + 2 - 2 = x$$

$$f(f^{-1}(x)) = \frac{1}{2}(2x - 2) + 1 = x - 1 + 1 = x.$$

See Figure 7.3.

EXAMPLE 3 Finding an Inverse Function

Find the inverse of the function $y = x^2, x \geq 0$, expressed as a function of x .

Solution We first solve for x in terms of y :

$$y = x^2$$

$$\sqrt{y} = \sqrt{x^2} = |x| = x \quad |x| = x \text{ because } x \geq 0$$

We then interchange x and y , obtaining

$$y = \sqrt{x}.$$

The inverse of the function $y = x^2, x \geq 0$, is the function $y = \sqrt{x}$ (Figure 7.4).

Notice that, unlike the restricted function $y = x^2, x \geq 0$, the unrestricted function $y = x^2$ is not one-to-one and therefore has no inverse.

Derivatives of Inverses of Differentiable Functions

If we calculate the derivatives of $f(x) = (1/2)x + 1$ and its inverse $f^{-1}(x) = 2x - 2$ from Example 2, we see that

$$\frac{d}{dx} f(x) = \frac{d}{dx} \left(\frac{1}{2}x + 1 \right) = \frac{1}{2}$$

$$\frac{d}{dx} f^{-1}(x) = \frac{d}{dx} (2x - 2) = 2.$$

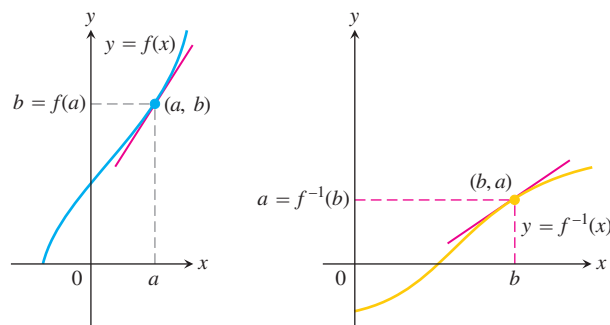
The derivatives are reciprocals of one another. The graph of f is the line $y = (1/2)x + 1$, and the graph of f^{-1} is the line $y = 2x - 2$ (Figure 7.3). Their slopes are reciprocals of one another.

This is not a special case. Reflecting any nonhorizontal or nonvertical line across the line $y = x$ always inverts the line's slope. If the original line has slope $m \neq 0$ (Figure 7.5), the reflected line has slope $1/m$ (Exercise 36).

The reciprocal relationship between the slopes of f and f^{-1} holds for other functions as well, but we must be careful to compare slopes at corresponding points. If the slope of $y = f(x)$ at the point $(a, f(a))$ is $f'(a)$ and $f'(a) \neq 0$, then the slope of $y = f^{-1}(x)$ at the point $(f(a), a)$ is the reciprocal $1/f'(a)$ (Figure 7.6). If we set $b = f(a)$, then

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}.$$

If $y = f(x)$ has a horizontal tangent line at $(a, f(a))$ then the inverse function f^{-1} has a vertical tangent line at $(f(a), a)$, and this infinite slope implies that f^{-1} is not differentiable



The slopes are reciprocal: $(f^{-1})'(b) = \frac{1}{f'(a)}$ or $(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$

FIGURE 7.6 The graphs of inverse functions have reciprocal slopes at corresponding points.

at $f(a)$. Theorem 1 gives the conditions under which f^{-1} is differentiable in its domain, which is the same as the range of f .

THEOREM 1 The Derivative Rule for Inverses

If f has an interval I as domain and $f'(x)$ exists and is never zero on I , then f^{-1} is differentiable at every point in its domain. The value of $(f^{-1})'$ at a point b in the domain of f^{-1} is the reciprocal of the value of f' at the point $a = f^{-1}(b)$:

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$$

or

$$\left. \frac{df^{-1}}{dx} \right|_{x=b} = \frac{1}{\left. \frac{df}{dx} \right|_{x=f^{-1}(b)}} \quad (1)$$

The proof of Theorem 1 is omitted, but here is another way to view it. When $y = f(x)$ is differentiable at $x = a$ and we change x by a small amount dx , the corresponding change in y is approximately

$$dy = f'(a) dx.$$

This means that y changes about $f'(a)$ times as fast as x when $x = a$ and that x changes about $1/f'(a)$ times as fast as y when $y = b$. It is reasonable that the derivative of f^{-1} at b is the reciprocal of the derivative of f at a .

EXAMPLE 4 Applying Theorem 1

The function $f(x) = x^2$, $x \geq 0$ and its inverse $f^{-1}(x) = \sqrt{x}$ have derivatives $f'(x) = 2x$ and $(f^{-1})'(x) = 1/(2\sqrt{x})$.

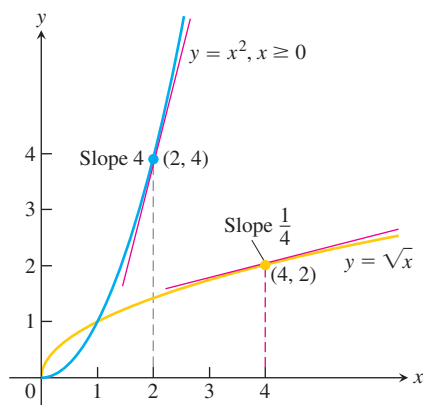


FIGURE 7.7 The derivative of $f^{-1}(x) = \sqrt{x}$ at the point $(4, 2)$ is the reciprocal of the derivative of $f(x) = x^2$ at $(2, 4)$ (Example 4).

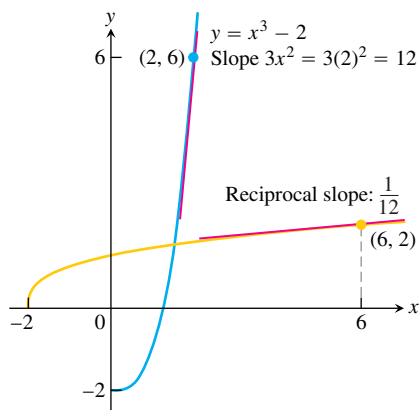


FIGURE 7.8 The derivative of $f(x) = x^3 - 2$ at $x = 2$ tells us the derivative of f^{-1} at $x = 6$ (Example 5).

Theorem 1 predicts that the derivative of $f^{-1}(x)$ is

$$\begin{aligned}(f^{-1})'(x) &= \frac{1}{f'(f^{-1}(x))} \\ &= \frac{1}{2(f^{-1}(x))} \\ &= \frac{1}{2(\sqrt{x})}.\end{aligned}$$

Theorem 1 gives a derivative that agrees with our calculation using the Power Rule for the derivative of the square root function.

Let's examine Theorem 1 at a specific point. We pick $x = 2$ (the number a) and $f(2) = 4$ (the value b). Theorem 1 says that the derivative of f at 2, $f'(2) = 4$, and the derivative of f^{-1} at $f(2)$, $(f^{-1})'(4)$, are reciprocals. It states that

$$(f^{-1})'(4) = \frac{1}{f'(f^{-1}(4))} = \frac{1}{f'(2)} = \frac{1}{2x}\bigg|_{x=2} = \frac{1}{4}.$$

See Figure 7.7.

Equation (1) sometimes enables us to find specific values of df^{-1}/dx without knowing a formula for f^{-1} .

EXAMPLE 5 Finding a Value of the Inverse Derivative

Let $f(x) = x^3 - 2$. Find the value of df^{-1}/dx at $x = 6 = f(2)$ without finding a formula for $f^{-1}(x)$.

Solution

$$\begin{aligned}\frac{df}{dx}\bigg|_{x=2} &= 3x^2\bigg|_{x=2} = 12 \\ \frac{df^{-1}}{dx}\bigg|_{x=f(2)} &= \frac{1}{\frac{df}{dx}\bigg|_{x=2}} = \frac{1}{12} \quad \text{Eq. (1)}\end{aligned}$$

See Figure 7.8.

Parametrizing Inverse Functions

We can graph or represent any function $y = f(x)$ parametrically as

$$x = t \quad \text{and} \quad y = f(t).$$

Interchanging t and $f(t)$ produces parametric equations for the inverse:

$$x = f(t) \quad \text{and} \quad y = t$$

(see Section 3.5).

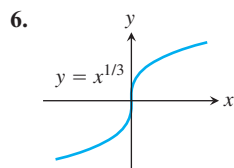
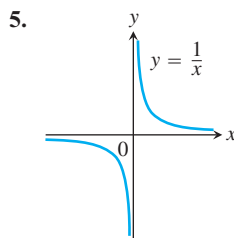
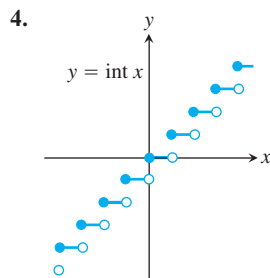
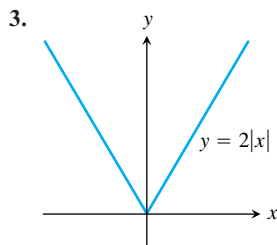
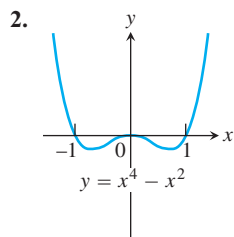
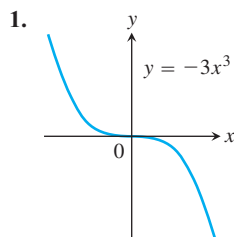
For example, to graph the one-to-one function $f(x) = x^2, x \geq 0$, on a grapher together with its inverse and the line $y = x, x \geq 0$, use the parametric graphing option with

$$\begin{aligned}\text{Graph of } f: & \quad x_1 = t, \quad y_1 = t^2, \quad t \geq 0 \\ \text{Graph of } f^{-1}: & \quad x_2 = t^2, \quad y_2 = t \\ \text{Graph of } y = x: & \quad x_3 = t, \quad y_3 = t\end{aligned}$$

EXERCISES 7.1

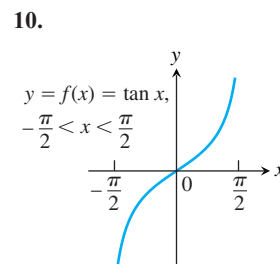
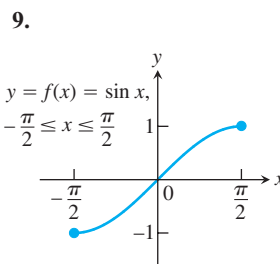
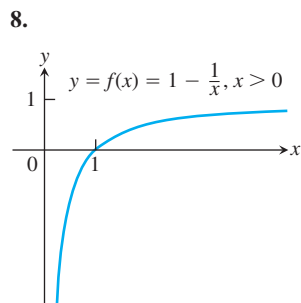
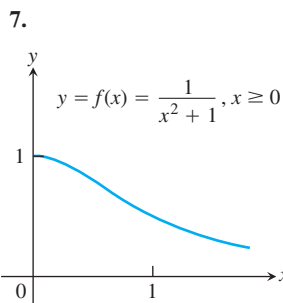
Identifying One-to-One Functions Graphically

Which of the functions graphed in Exercises 1–6 are one-to-one, and which are not?



Graphing Inverse Functions

Each of Exercises 7–10 shows the graph of a function $y = f(x)$. Copy the graph and draw in the line $y = x$. Then use symmetry with respect to the line $y = x$ to add the graph of f^{-1} to your sketch. (It is not necessary to find a formula for f^{-1} .) Identify the domain and range of f^{-1} .



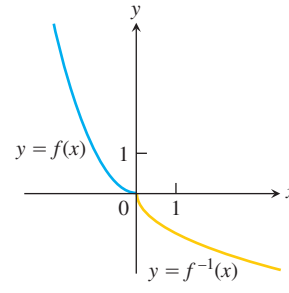
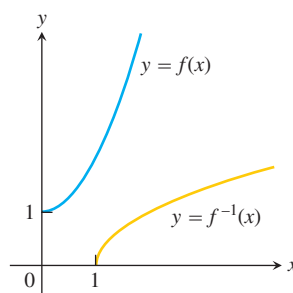
11. a. Graph the function $f(x) = \sqrt{1 - x^2}, 0 \leq x \leq 1$. What symmetry does the graph have?
 b. Show that f is its own inverse. (Remember that $\sqrt{x^2} = x$ if $x \geq 0$.)
12. a. Graph the function $f(x) = 1/x$. What symmetry does the graph have?
 b. Show that f is its own inverse.

Formulas for Inverse Functions

Each of Exercises 13–18 gives a formula for a function $y = f(x)$ and shows the graphs of f and f^{-1} . Find a formula for f^{-1} in each case.

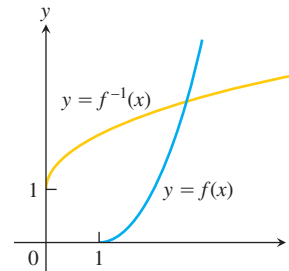
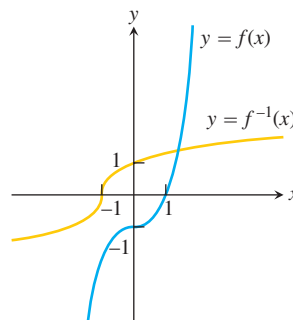
13. $f(x) = x^2 + 1, x \geq 0$

14. $f(x) = x^2, x \leq 0$

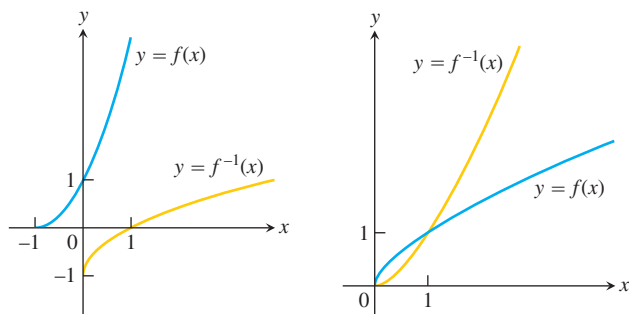


15. $f(x) = x^3 - 1$

16. $f(x) = x^2 - 2x + 1, x \geq 1$



17. $f(x) = (x + 1)^2, x \geq -1$ 18. $f(x) = x^{2/3}, x \geq 0$



Each of Exercises 19–24 gives a formula for a function $y = f(x)$. In each case, find $f^{-1}(x)$ and identify the domain and range of f^{-1} . As a check, show that $f(f^{-1}(x)) = f^{-1}(f(x)) = x$.

19. $f(x) = x^5$ 20. $f(x) = x^4, x \geq 0$
 21. $f(x) = x^3 + 1$ 22. $f(x) = (1/2)x - 7/2$
 23. $f(x) = 1/x^2, x > 0$ 24. $f(x) = 1/x^3, x \neq 0$

Derivatives of Inverse Functions

In Exercises 25–28:

- Find $f^{-1}(x)$.
 - Graph f and f^{-1} together.
 - Evaluate df/dx at $x = a$ and df^{-1}/dx at $x = f(a)$ to show that at these points $df^{-1}/dx = 1/(df/dx)$.
25. $f(x) = 2x + 3, a = -1$ 26. $f(x) = (1/5)x + 7, a = -1$
 27. $f(x) = 5 - 4x, a = 1/2$ 28. $f(x) = 2x^2, x \geq 0, a = 5$
 29. a. Show that $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ are inverses of one another.
 b. Graph f and g over an x -interval large enough to show the graphs intersecting at $(1, 1)$ and $(-1, -1)$. Be sure the picture shows the required symmetry about the line $y = x$.
 c. Find the slopes of the tangents to the graphs of f and g at $(1, 1)$ and $(-1, -1)$ (four tangents in all).
 d. What lines are tangent to the curves at the origin?
30. a. Show that $h(x) = x^3/4$ and $k(x) = (4x)^{1/3}$ are inverses of one another.
 b. Graph h and k over an x -interval large enough to show the graphs intersecting at $(2, 2)$ and $(-2, -2)$. Be sure the picture shows the required symmetry about the line $y = x$.
 c. Find the slopes of the tangents to the graphs at h and k at $(2, 2)$ and $(-2, -2)$.
 d. What lines are tangent to the curves at the origin?
31. Let $f(x) = x^3 - 3x^2 - 1, x \geq 2$. Find the value of df^{-1}/dx at the point $x = -1 = f(3)$.
 32. Let $f(x) = x^2 - 4x - 5, x > 2$. Find the value of df^{-1}/dx at the point $x = 0 = f(5)$.

33. Suppose that the differentiable function $y = f(x)$ has an inverse and that the graph of f passes through the point $(2, 4)$ and has a slope of $1/3$ there. Find the value of df^{-1}/dx at $x = 4$.
 34. Suppose that the differentiable function $y = g(x)$ has an inverse and that the graph of g passes through the origin with slope 2. Find the slope of the graph of g^{-1} at the origin.

Inverses of Lines

35. a. Find the inverse of the function $f(x) = mx$, where m is a constant different from zero.
 b. What can you conclude about the inverse of a function $y = f(x)$ whose graph is a line through the origin with a nonzero slope m ?
36. Show that the graph of the inverse of $f(x) = mx + b$, where m and b are constants and $m \neq 0$, is a line with slope $1/m$ and y -intercept $-b/m$.
37. a. Find the inverse of $f(x) = x + 1$. Graph f and its inverse together. Add the line $y = x$ to your sketch, drawing it with dashes or dots for contrast.
 b. Find the inverse of $f(x) = x + b$ (b constant). How is the graph of f^{-1} related to the graph of f ?
 c. What can you conclude about the inverses of functions whose graphs are lines parallel to the line $y = x$?
38. a. Find the inverse of $f(x) = -x + 1$. Graph the line $y = -x + 1$ together with the line $y = x$. At what angle do the lines intersect?
 b. Find the inverse of $f(x) = -x + b$ (b constant). What angle does the line $y = -x + b$ make with the line $y = x$?
 c. What can you conclude about the inverses of functions whose graphs are lines perpendicular to the line $y = x$?

Increasing and Decreasing Functions

39. As in Section 4.3, a function $f(x)$ increases on an interval I if for any two points x_1 and x_2 in I ,

$$x_2 > x_1 \implies f(x_2) > f(x_1).$$

Similarly, a function decreases on I if for any two points x_1 and x_2 in I ,

$$x_2 > x_1 \implies f(x_2) < f(x_1).$$

Show that increasing functions and decreasing functions are one-to-one. That is, show that for any x_1 and x_2 in I , $x_2 \neq x_1$ implies $f(x_2) \neq f(x_1)$.

Use the results of Exercise 39 to show that the functions in Exercises 40–44 have inverses over their domains. Find a formula for df^{-1}/dx using Theorem 1.

40. $f(x) = (1/3)x + (5/6)$ 41. $f(x) = 27x^3$
 42. $f(x) = 1 - 8x^3$ 43. $f(x) = (1 - x)^3$
 44. $f(x) = x^{5/3}$

Theory and Applications

45. If $f(x)$ is one-to-one, can anything be said about $g(x) = -f(x)$? Is it also one-to-one? Give reasons for your answer.
46. If $f(x)$ is one-to-one and $f(x)$ is never zero, can anything be said about $h(x) = 1/f(x)$? Is it also one-to-one? Give reasons for your answer.
47. Suppose that the range of g lies in the domain of f so that the composite $f \circ g$ is defined. If f and g are one-to-one, can anything be said about $f \circ g$? Give reasons for your answer.
48. If a composite $f \circ g$ is one-to-one, must g be one-to-one? Give reasons for your answer.
49. Suppose $f(x)$ is positive, continuous, and increasing over the interval $[a, b]$. By interpreting the graph of f show that
- $$\int_a^b f(x) dx + \int_{f(a)}^{f(b)} f^{-1}(y) dy = bf(b) - af(a).$$
50. Determine conditions on the constants a, b, c , and d so that the rational function

$$f(x) = \frac{ax + b}{cx + d}$$

has an inverse.

51. If we write $g(x)$ for $f^{-1}(x)$, Equation (1) can be written as

$$g'(f(a)) = \frac{1}{f'(a)}, \quad \text{or} \quad g'(f(a)) \cdot f'(a) = 1.$$

If we then write x for a , we get

$$g'(f(x)) \cdot f'(x) = 1.$$

The latter equation may remind you of the Chain Rule, and indeed there is a connection.

Assume that f and g are differentiable functions that are inverses of one another, so that $(g \circ f)(x) = x$. Differentiate both sides of this equation with respect to x , using the Chain Rule to express $(g \circ f)'(x)$ as a product of derivatives of g and f . What do you find? (This is not a proof of Theorem 1 because we assume here the theorem's conclusion that $g = f^{-1}$ is differentiable.)

52. **Equivalence of the washer and shell methods for finding volume**
Let f be differentiable and increasing on the interval $a \leq x \leq b$, with $a > 0$, and suppose that f has a differentiable inverse, f^{-1} . Revolve about the y -axis the region bounded by the graph of f and the lines $x = a$ and $y = f(b)$ to generate a solid. Then the values of the integrals given by the washer and shell methods for the volume have identical values:

$$\int_{f(a)}^{f(b)} \pi((f^{-1}(y))^2 - a^2) dy = \int_a^b 2\pi x(f(b) - f(x)) dx.$$

To prove this equality, define

$$W(t) = \int_{f(a)}^{f(t)} \pi((f^{-1}(y))^2 - a^2) dy$$

$$S(t) = \int_a^t 2\pi x(f(t) - f(x)) dx.$$

Then show that the functions W and S agree at a point of $[a, b]$ and have identical derivatives on $[a, b]$. As you saw in Section 4.8, Exercise 102, this will guarantee $W(t) = S(t)$ for all t in $[a, b]$. In particular, $W(b) = S(b)$. (Source: "Disks and Shells Revisited," by Walter Carlip, *American Mathematical Monthly*, Vol. 98, No. 2, Feb. 1991, pp. 154–156.)

COMPUTER EXPLORATIONS

In Exercises 53–60, you will explore some functions and their inverses together with their derivatives and linear approximating functions at specified points. Perform the following steps using your CAS:

- Plot the function $y = f(x)$ together with its derivative over the given interval. Explain why you know that f is one-to-one over the interval.
- Solve the equation $y = f(x)$ for x as a function of y , and name the resulting inverse function g .
- Find the equation for the tangent line to f at the specified point $(x_0, f(x_0))$.
- Find the equation for the tangent line to g at the point $(f(x_0), x_0)$ located symmetrically across the 45° line $y = x$ (which is the graph of the identity function). Use Theorem 1 to find the slope of this tangent line.
- Plot the functions f and g , the identity, the two tangent lines, and the line segment joining the points $(x_0, f(x_0))$ and $(f(x_0), x_0)$. Discuss the symmetries you see across the main diagonal.

53. $y = \sqrt{3x - 2}$, $\frac{2}{3} \leq x \leq 4$, $x_0 = 3$

54. $y = \frac{3x + 2}{2x - 11}$, $-2 \leq x \leq 2$, $x_0 = 1/2$

55. $y = \frac{4x}{x^2 + 1}$, $-1 \leq x \leq 1$, $x_0 = 1/2$

56. $y = \frac{x^3}{x^2 + 1}$, $-1 \leq x \leq 1$, $x_0 = 1/2$

57. $y = x^3 - 3x^2 - 1$, $2 \leq x \leq 5$, $x_0 = \frac{27}{10}$

58. $y = 2 - x - x^3$, $-2 \leq x \leq 2$, $x_0 = \frac{3}{2}$

59. $y = e^x$, $-3 \leq x \leq 5$, $x_0 = 1$

60. $y = \sin x$, $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, $x_0 = 1$

In Exercises 61 and 62, repeat the steps above to solve for the functions $y = f(x)$ and $x = f^{-1}(y)$ defined implicitly by the given equations over the interval.

61. $y^{1/3} - 1 = (x + 2)^3$, $-5 \leq x \leq 5$, $x_0 = -3/2$

62. $\cos y = x^{1/5}$, $0 \leq x \leq 1$, $x_0 = 1/2$

7.2

Natural Logarithms

For any positive number a , the function value $f(x) = a^x$ is easy to define when x is an integer or rational number. When x is irrational, the meaning of a^x is not so clear. Similarly, the definition of the logarithm $\log_a x$, the inverse function of $f(x) = a^x$, is not completely obvious. In this section we use integral calculus to define the *natural logarithm* function, for which the number a is a particularly important value. This function allows us to define and analyze general exponential and logarithmic functions, $y = a^x$ and $y = \log_a x$.

Logarithms originally played important roles in arithmetic computations. Historically, considerable labor went into producing long tables of logarithms, correct to five, eight, or even more, decimal places of accuracy. Prior to the modern age of electronic calculators and computers, every engineer owned slide rules marked with logarithmic scales. Calculations with logarithms made possible the great seventeenth-century advances in offshore navigation and celestial mechanics. Today we know such calculations are done using calculators or computers, but the properties and numerous applications of logarithms are as important as ever.

Definition of the Natural Logarithm Function

One solid approach to defining and understanding logarithms begins with a study of the natural logarithm function defined as an integral through the Fundamental Theorem of Calculus. While this approach may seem indirect, it enables us to derive quickly the familiar properties of logarithmic and exponential functions. The functions we have studied so far were analyzed using the techniques of calculus, but here we do something more fundamental. We use calculus for the very definition of the logarithmic and exponential functions.

The natural logarithm of a positive number x , written as $\ln x$, is the value of an integral.

DEFINITION The Natural Logarithm Function

$$\ln x = \int_1^x \frac{1}{t} dt, \quad x > 0$$

If $x > 1$, then $\ln x$ is the area under the curve $y = 1/t$ from $t = 1$ to $t = x$ (Figure 7.9). For $0 < x < 1$, $\ln x$ gives the negative of the area under the curve from x to 1. The function is not defined for $x \leq 0$. From the Zero Width Interval Rule for definite integrals, we also have

$$\ln 1 = \int_1^1 \frac{1}{t} dt = 0.$$

The Derivative of $y = \ln x$

By the first part of the Fundamental Theorem of Calculus (Section 5.4),

$$\frac{d}{dx} \ln x = \frac{d}{dx} \int_1^x \frac{1}{t} dt = \frac{1}{x}.$$

For every positive value of x , we have

$$\frac{d}{dx} \ln x = \frac{1}{x}.$$

Therefore, the function $y = \ln x$ is a solution to the initial value problem $dy/dx = 1/x$, $x > 0$, with $y(1) = 0$. Notice that the derivative is always positive so the natural logarithm is an increasing function, hence it is one-to-one and invertible. Its inverse is studied in Section 7.3.

If u is a differentiable function of x whose values are positive, so that $\ln u$ is defined, then applying the Chain Rule

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

to the function $y = \ln u$ gives

$$\frac{d}{dx} \ln u = \frac{d}{du} \ln u \cdot \frac{du}{dx} = \frac{1}{u} \frac{du}{dx}.$$

$$\frac{d}{dx} \ln u = \frac{1}{u} \frac{du}{dx}, \quad u > 0 \quad (1)$$

EXAMPLE 1 Derivatives of Natural Logarithms

(a) $\frac{d}{dx} \ln 2x = \frac{1}{2x} \frac{d}{dx} (2x) = \frac{1}{2x} (2) = \frac{1}{x}$

(b) Equation (1) with $u = x^2 + 3$ gives

$$\frac{d}{dx} \ln(x^2 + 3) = \frac{1}{x^2 + 3} \cdot \frac{d}{dx} (x^2 + 3) = \frac{1}{x^2 + 3} \cdot 2x = \frac{2x}{x^2 + 3}.$$

Notice the remarkable occurrence in Example 1a. The function $y = \ln 2x$ has the same derivative as the function $y = \ln x$. This is true of $y = \ln ax$ for any positive number a :

$$\frac{d}{dx} \ln ax = \frac{1}{ax} \cdot \frac{d}{dx} (ax) = \frac{1}{ax} (a) = \frac{1}{x}. \quad (2)$$

Since they have the same derivative, the functions $y = \ln ax$ and $y = \ln x$ differ by a constant.

HISTORICAL BIOGRAPHY

John Napier
(1550–1617)

Properties of Logarithms

Logarithms were invented by John Napier and were the single most important improvement in arithmetic calculation before the modern electronic computer. What made them so useful is that the properties of logarithms enable multiplication of positive numbers by addition of their logarithms, division of positive numbers by subtraction of their logarithms, and exponentiation of a number by multiplying its logarithm by the exponent. We summarize these properties as a series of rules in Theorem 2. For the moment, we restrict the exponent r in Rule 4 to be a rational number; you will see why when we prove the rule.

THEOREM 2 Properties of Logarithms

For any numbers $a > 0$ and $x > 0$, the natural logarithm satisfies the following rules:

- | | | |
|----------------------------|-----------------------------------|---------------------|
| 1. <i>Product Rule:</i> | $\ln ax = \ln a + \ln x$ | |
| 2. <i>Quotient Rule:</i> | $\ln \frac{a}{x} = \ln a - \ln x$ | |
| 3. <i>Reciprocal Rule:</i> | $\ln \frac{1}{x} = -\ln x$ | Rule 2 with $a = 1$ |
| 4. <i>Power Rule:</i> | $\ln x^r = r \ln x$ | r rational |

We illustrate how these rules apply.

EXAMPLE 2 Interpreting the Properties of Logarithms

- (a) $\ln 6 = \ln (2 \cdot 3) = \ln 2 + \ln 3$ Product
- (b) $\ln 4 - \ln 5 = \ln \frac{4}{5} = \ln 0.8$ Quotient
- (c) $\ln \frac{1}{8} = -\ln 8$ Reciprocal
- $= -\ln 2^3 = -3 \ln 2$ Power

EXAMPLE 3 Applying the Properties to Function Formulas

- (a) $\ln 4 + \ln \sin x = \ln (4 \sin x)$ Product
- (b) $\ln \frac{x+1}{2x-3} = \ln (x+1) - \ln (2x-3)$ Quotient

$$(c) \ln \sec x = \ln \frac{1}{\cos x} = -\ln \cos x \quad \text{Reciprocal}$$

$$(d) \ln \sqrt[3]{x+1} = \ln (x+1)^{1/3} = \frac{1}{3} \ln (x+1) \quad \text{Power} \quad \blacksquare$$

We now give the proof of Theorem 2. The steps in the proof are similar to those used in solving problems involving logarithms.

Proof that $\ln ax = \ln a + \ln x$ The argument is unusual—and elegant. It starts by observing that $\ln ax$ and $\ln x$ have the same derivative (Equation 2). According to Corollary 2 of the Mean Value Theorem, then, the functions must differ by a constant, which means that

$$\ln ax = \ln x + C$$

for some C .

Since this last equation holds for all positive values of x , it must hold for $x = 1$. Hence,

$$\begin{aligned} \ln(a \cdot 1) &= \ln 1 + C \\ \ln a &= 0 + C && \ln 1 = 0 \\ C &= \ln a. \end{aligned}$$

By substituting we conclude,

$$\ln ax = \ln a + \ln x.$$

Proof that $\ln x^r = r \ln x$ (assuming r rational) We use the same-derivative argument again. For all positive values of x ,

$$\begin{aligned} \frac{d}{dx} \ln x^r &= \frac{1}{x^r} \frac{d}{dx} (x^r) && \text{Eq. (1) with } u = x^r \\ &= \frac{1}{x^r} r x^{r-1} && \text{Here is where we need } r \text{ to be rational,} \\ &= r \cdot \frac{1}{x} = \frac{d}{dx} (r \ln x). && \text{at least for now. We have proved the} \\ &&& \text{Power Rule only for rational} \\ &&& \text{exponents.} \end{aligned}$$

Since $\ln x^r$ and $r \ln x$ have the same derivative,

$$\ln x^r = r \ln x + C$$

for some constant C . Taking x to be 1 identifies C as zero, and we're done.

You are asked to prove Rule 2 in Exercise 84. Rule 3 is a special case of Rule 2, obtained by setting $a = 1$ and noting that $\ln 1 = 0$. So we have established all cases of Theorem 2. ■

We have not yet proved Rule 4 for r irrational; we will return to this case in Section 7.3. The rule does hold for all r , rational or irrational.

The Graph and Range of $\ln x$

The derivative $d(\ln x)/dx = 1/x$ is positive for $x > 0$, so $\ln x$ is an increasing function of x . The second derivative, $-1/x^2$, is negative, so the graph of $\ln x$ is concave down.

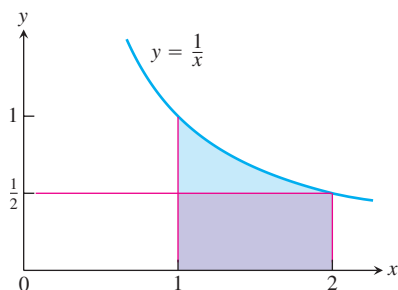


FIGURE 7.10 The rectangle of height $y = 1/2$ fits beneath the graph of $y = 1/x$ for the interval $1 \leq x \leq 2$.

We can estimate the value of $\ln 2$ by considering the area under the graph of $y = 1/x$ and above the interval $[1, 2]$. In Figure 7.10 a rectangle of height $1/2$ over the interval $[1, 2]$ fits under the graph. Therefore the area under the graph, which is $\ln 2$, is greater than the area, $1/2$, of the rectangle. So $\ln 2 > 1/2$. Knowing this we have,

$$\ln 2^n = n \ln 2 > n \left(\frac{1}{2} \right) = \frac{n}{2}$$

and

$$\ln 2^{-n} = -n \ln 2 < -n \left(\frac{1}{2} \right) = -\frac{n}{2}.$$

It follows that

$$\lim_{x \rightarrow \infty} \ln x = \infty \quad \text{and} \quad \lim_{x \rightarrow 0^+} \ln x = -\infty.$$

We defined $\ln x$ for $x > 0$, so the domain of $\ln x$ is the set of positive real numbers. The above discussion and the Intermediate Value Theorem show that its range is the entire real line giving the graph of $y = \ln x$ shown in Figure 7.9.

The Integral $\int (1/u) du$

Equation (1) leads to the integral formula

$$\int \frac{1}{u} du = \ln u + C \quad (3)$$

when u is a positive differentiable function, but what if u is negative? If u is negative, then $-u$ is positive and

$$\begin{aligned} \int \frac{1}{u} du &= \int \frac{1}{(-u)} d(-u) && \text{Eq. (3) with } u \text{ replaced by } -u \\ &= \ln(-u) + C. \end{aligned} \quad (4)$$

We can combine Equations (3) and (4) into a single formula by noticing that in each case the expression on the right is $\ln |u| + C$. In Equation (3), $\ln u = \ln |u|$ because $u > 0$; in Equation (4), $\ln(-u) = \ln |u|$ because $u < 0$. Whether u is positive or negative, the integral of $(1/u) du$ is $\ln |u| + C$.

If u is a differentiable function that is never zero,

$$\int \frac{1}{u} du = \ln |u| + C. \quad (5)$$

Equation (5) applies anywhere on the domain of $1/u$, the points where $u \neq 0$.

We know that

$$\int u^n du = \frac{u^{n+1}}{n+1} + C, \quad n \neq -1 \text{ and rational}$$

Equation (5) explains what to do when n equals -1 . Equation (5) says integrals of a certain form lead to logarithms. If $u = f(x)$, then $du = f'(x) dx$ and

$$\int \frac{1}{u} du = \int \frac{f'(x)}{f(x)} dx.$$

So Equation (5) gives

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

whenever $f(x)$ is a differentiable function that maintains a constant sign on the domain given for it.

EXAMPLE 4 Applying Equation (5)

$$\begin{aligned} \text{(a)} \quad \int_0^2 \frac{2x}{x^2 - 5} dx &= \int_{-5}^{-1} \frac{du}{u} = \ln |u| \Big|_{-5}^{-1} & u = x^2 - 5, \quad du = 2x dx, \\ & & u(0) = -5, \quad u(2) = -1 \\ &= \ln |-1| - \ln |-5| = \ln 1 - \ln 5 = -\ln 5 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int_{-\pi/2}^{\pi/2} \frac{4 \cos \theta}{3 + 2 \sin \theta} d\theta &= \int_1^5 \frac{2}{u} du & u = 3 + 2 \sin \theta, \quad du = 2 \cos \theta d\theta, \\ & & u(-\pi/2) = 1, \quad u(\pi/2) = 5 \\ &= 2 \ln |u| \Big|_1^5 \\ &= 2 \ln |5| - 2 \ln |1| = 2 \ln 5 \end{aligned}$$

Note that $u = 3 + 2 \sin \theta$ is always positive on $[-\pi/2, \pi/2]$, so Equation (5) applies. ■

The Integrals of $\tan x$ and $\cot x$

Equation (5) tells us at last how to integrate the tangent and cotangent functions. For the tangent function,

$$\begin{aligned} \int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{-du}{u} & u = \cos x > 0 \text{ on } (-\pi/2, \pi/2), \\ & & du = -\sin x dx \\ &= -\int \frac{du}{u} = -\ln |u| + C \\ &= -\ln |\cos x| + C = \ln \frac{1}{|\cos x|} + C & \text{Reciprocal Rule} \\ &= \ln |\sec x| + C. \end{aligned}$$

For the cotangent,

$$\begin{aligned} \int \cot x dx &= \int \frac{\cos x}{\sin x} dx = \int \frac{du}{u} & u = \sin x, \\ & & du = \cos x dx \\ &= \ln |u| + C = \ln |\sin x| + C = -\ln |\csc x| + C. \end{aligned}$$

$$\int \tan u \, du = -\ln |\cos u| + C = \ln |\sec u| + C$$

$$\int \cot u \, du = \ln |\sin u| + C = -\ln |\csc u| + C$$

EXAMPLE 5

$$\begin{aligned} \int_0^{\pi/6} \tan 2x \, dx &= \int_0^{\pi/3} \tan u \cdot \frac{du}{2} = \frac{1}{2} \int_0^{\pi/3} \tan u \, du \\ &= \frac{1}{2} \ln |\sec u| \Big|_0^{\pi/3} = \frac{1}{2} (\ln 2 - \ln 1) = \frac{1}{2} \ln 2 \end{aligned}$$

Substitute $u = 2x$,
 $dx = du/2$,
 $u(0) = 0$,
 $u(\pi/6) = \pi/3$

Logarithmic Differentiation

The derivatives of positive functions given by formulas that involve products, quotients, and powers can often be found more quickly if we take the natural logarithm of both sides before differentiating. This enables us to use the laws of logarithms to simplify the formulas before differentiating. The process, called **logarithmic differentiation**, is illustrated in the next example.

EXAMPLE 6 Using Logarithmic Differentiation

Find dy/dx if

$$y = \frac{(x^2 + 1)(x + 3)^{1/2}}{x - 1}, \quad x > 1.$$

Solution We take the natural logarithm of both sides and simplify the result with the properties of logarithms:

$$\begin{aligned} \ln y &= \ln \frac{(x^2 + 1)(x + 3)^{1/2}}{x - 1} \\ &= \ln((x^2 + 1)(x + 3)^{1/2}) - \ln(x - 1) && \text{Rule 2} \\ &= \ln(x^2 + 1) + \ln(x + 3)^{1/2} - \ln(x - 1) && \text{Rule 1} \\ &= \ln(x^2 + 1) + \frac{1}{2} \ln(x + 3) - \ln(x - 1). && \text{Rule 3} \end{aligned}$$

We then take derivatives of both sides with respect to x , using Equation (1) on the left:

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x^2 + 1} \cdot 2x + \frac{1}{2} \cdot \frac{1}{x + 3} - \frac{1}{x - 1}.$$

Next we solve for dy/dx :

$$\frac{dy}{dx} = y \left(\frac{2x}{x^2 + 1} + \frac{1}{2x + 6} - \frac{1}{x - 1} \right).$$

Finally, we substitute for y :

$$\frac{dy}{dx} = \frac{(x^2 + 1)(x + 3)^{1/2}}{x - 1} \left(\frac{2x}{x^2 + 1} + \frac{1}{2x + 6} - \frac{1}{x - 1} \right).$$

A direct computation in Example 6, using the Quotient and Product Rules, would be much longer. ■

EXERCISES 7.2

Using the Properties of Logarithms

- Express the following logarithms in terms of $\ln 2$ and $\ln 3$.
 - $\ln 0.75$
 - $\ln(4/9)$
 - $\ln(1/2)$
 - $\ln \sqrt[3]{9}$
 - $\ln 3\sqrt{2}$
 - $\ln \sqrt{13.5}$
- Express the following logarithms in terms of $\ln 5$ and $\ln 7$.
 - $\ln(1/125)$
 - $\ln 9.8$
 - $\ln 7\sqrt{7}$
 - $\ln 1225$
 - $\ln 0.056$
 - $(\ln 35 + \ln(1/7))/(\ln 25)$

Use the properties of logarithms to simplify the expressions in Exercises 3 and 4.

- $\ln \sin \theta - \ln \left(\frac{\sin \theta}{5} \right)$
 - $\ln(3x^2 - 9x) + \ln \left(\frac{1}{3x} \right)$
 - $\frac{1}{2} \ln(4t^4) - \ln 2$
- $\ln \sec \theta + \ln \cos \theta$
 - $\ln(8x + 4) - 2 \ln 2$
 - $3 \ln \sqrt[3]{t^2 - 1} - \ln(t + 1)$

Derivatives of Logarithms

In Exercises 5–36, find the derivative of y with respect to x , t , or θ , as appropriate.

- $y = \ln 3x$
- $y = \ln kx$, k constant
- $y = \ln(t^2)$
- $y = \ln(t^{3/2})$
- $y = \ln \frac{3}{x}$
- $y = \ln \frac{10}{x}$
- $y = \ln(\theta + 1)$
- $y = \ln(2\theta + 2)$
- $y = \ln x^3$
- $y = (\ln x)^3$
- $y = t(\ln t)^2$
- $y = t\sqrt{\ln t}$
- $y = \frac{x^4}{4} \ln x - \frac{x^4}{16}$
- $y = \frac{x^3}{3} \ln x - \frac{x^3}{9}$
- $y = \frac{\ln t}{t}$
- $y = \frac{1 + \ln t}{t}$
- $y = \frac{\ln x}{1 + \ln x}$
- $y = \frac{x \ln x}{1 + \ln x}$
- $y = \ln(\ln x)$
- $y = \ln(\ln(\ln x))$
- $y = \theta(\sin(\ln \theta) + \cos(\ln \theta))$
- $y = \ln(\sec \theta + \tan \theta)$
- $y = \ln \frac{1}{x\sqrt{x+1}}$
- $y = \frac{1}{2} \ln \frac{1+x}{1-x}$
- $y = \frac{1 + \ln t}{1 - \ln t}$
- $y = \sqrt{\ln \sqrt{t}}$
- $y = \ln(\sec(\ln \theta))$
- $y = \ln \left(\frac{\sqrt{\sin \theta \cos \theta}}{1 + 2 \ln \theta} \right)$
- $y = \ln \left(\frac{(x^2 + 1)^5}{\sqrt{1-x}} \right)$
- $y = \ln \sqrt{\frac{(x+1)^5}{(x+2)^{20}}}$
- $y = \int_{x^2/2}^{x^2} \ln \sqrt{t} \, dt$
- $y = \int_{\sqrt{x}}^{\sqrt[3]{x}} \ln t \, dt$

Integration

Evaluate the integrals in Exercises 37–54.

- $\int_{-3}^{-2} \frac{dx}{x}$
- $\int_{-1}^0 \frac{3 \, dx}{3x - 2}$
- $\int \frac{2y \, dy}{y^2 - 25}$
- $\int \frac{8r \, dr}{4r^2 - 5}$
- $\int_0^\pi \frac{\sin t}{2 - \cos t} \, dt$
- $\int_0^{\pi/3} \frac{4 \sin \theta}{1 - 4 \cos \theta} \, d\theta$
- $\int_1^2 \frac{2 \ln x}{x} \, dx$
- $\int_2^4 \frac{dx}{x \ln x}$
- $\int_2^4 \frac{dx}{x(\ln x)^2}$
- $\int_2^{16} \frac{dx}{2x\sqrt{\ln x}}$
- $\int \frac{3 \sec^2 t}{6 + 3 \tan t} \, dt$
- $\int \frac{\sec y \tan y}{2 + \sec y} \, dy$
- $\int_0^{\pi/2} \tan \frac{x}{2} \, dx$
- $\int_{\pi/2}^\pi 2 \cot \frac{\theta}{3} \, d\theta$
- $\int \frac{dx}{2\sqrt{x} + 2x}$
- $\int \frac{\sec x \, dx}{\sqrt{\ln(\sec x + \tan x)}}$
- $\int_{\pi/4}^{\pi/2} \cot t \, dt$
- $\int_0^{\pi/12} 6 \tan 3x \, dx$

Logarithmic Differentiation

In Exercises 55–68, use logarithmic differentiation to find the derivative of y with respect to the given independent variable.

55. $y = \sqrt{x(x+1)}$ 56. $y = \sqrt{(x^2+1)(x-1)^2}$
57. $y = \sqrt{\frac{t}{t+1}}$ 58. $y = \sqrt{\frac{1}{t(t+1)}}$
59. $y = \sqrt{\theta+3} \sin \theta$ 60. $y = (\tan \theta) \sqrt{2\theta+1}$
61. $y = t(t+1)(t+2)$ 62. $y = \frac{1}{t(t+1)(t+2)}$
63. $y = \frac{\theta+5}{\theta \cos \theta}$ 64. $y = \frac{\theta \sin \theta}{\sqrt{\sec \theta}}$
65. $y = \frac{x\sqrt{x^2+1}}{(x+1)^{2/3}}$ 66. $y = \sqrt{\frac{(x+1)^{10}}{(2x+1)^5}}$
67. $y = \sqrt[3]{\frac{x(x-2)}{x^2+1}}$ 68. $y = \sqrt[3]{\frac{x(x+1)(x-2)}{(x^2+1)(2x+3)}}$

Theory and Applications

69. Locate and identify the absolute extreme values of
- $\ln(\cos x)$ on $[-\pi/4, \pi/3]$,
 - $\cos(\ln x)$ on $[1/2, 2]$.
70. a. Prove that $f(x) = x - \ln x$ is increasing for $x > 1$.
b. Using part (a), show that $\ln x < x$ if $x > 1$.
71. Find the area between the curves $y = \ln x$ and $y = \ln 2x$ from $x = 1$ to $x = 5$.
72. Find the area between the curve $y = \tan x$ and the x -axis from $x = -\pi/4$ to $x = \pi/3$.
73. The region in the first quadrant bounded by the coordinate axes, the line $y = 3$, and the curve $x = 2/\sqrt{y+1}$ is revolved about the y -axis to generate a solid. Find the volume of the solid.
74. The region between the curve $y = \sqrt{\cot x}$ and the x -axis from $x = \pi/6$ to $x = \pi/2$ is revolved about the x -axis to generate a solid. Find the volume of the solid.
75. The region between the curve $y = 1/x^2$ and the x -axis from $x = 1/2$ to $x = 2$ is revolved about the y -axis to generate a solid. Find the volume of the solid.
76. In Section 6.2, Exercise 6, we revolved about the y -axis the region between the curve $y = 9x/\sqrt{x^3+9}$ and the x -axis from $x = 0$ to $x = 3$ to generate a solid of volume 36π . What volume do you get if you revolve the region about the x -axis instead? (See Section 6.2, Exercise 6, for a graph.)
77. Find the lengths of the following curves.
- $y = (x^2/8) - \ln x$, $4 \leq x \leq 8$
 - $x = (y/4)^2 - 2 \ln(y/4)$, $4 \leq y \leq 12$
78. Find a curve through the point $(1, 0)$ whose length from $x = 1$ to

$x = 2$ is

$$L = \int_1^2 \sqrt{1 + \frac{1}{x^2}} dx.$$

- T** 79. a. Find the centroid of the region between the curve $y = 1/x$ and the x -axis from $x = 1$ to $x = 2$. Give the coordinates to two decimal places.
b. Sketch the region and show the centroid in your sketch.
80. a. Find the center of mass of a thin plate of constant density covering the region between the curve $y = 1/\sqrt{x}$ and the x -axis from $x = 1$ to $x = 16$.
b. Find the center of mass if, instead of being constant, the density function is $\delta(x) = 4/\sqrt{x}$.
- Solve the initial value problems in Exercises 81 and 82.
81. $\frac{dy}{dx} = 1 + \frac{1}{x}$, $y(1) = 3$
82. $\frac{d^2y}{dx^2} = \sec^2 x$, $y(0) = 0$ and $y'(0) = 1$
- T** 83. **The linearization of $\ln(1+x)$ at $x = 0$** Instead of approximating $\ln x$ near $x = 1$, we approximate $\ln(1+x)$ near $x = 0$. We get a simpler formula this way.
- Derive the linearization $\ln(1+x) \approx x$ at $x = 0$.
 - Estimate to five decimal places the error involved in replacing $\ln(1+x)$ by x on the interval $[0, 0.1]$.
 - Graph $\ln(1+x)$ and x together for $0 \leq x \leq 0.5$. Use different colors, if available. At what points does the approximation of $\ln(1+x)$ seem best? Least good? By reading coordinates from the graphs, find as good an upper bound for the error as your grapher will allow.
84. Use the same-derivative argument, as was done to prove Rules 1 and 4 of Theorem 2, to prove the Quotient Rule property of logarithms.

Grapher Explorations

85. Graph $\ln x$, $\ln 2x$, $\ln 4x$, $\ln 8x$, and $\ln 16x$ (as many as you can) together for $0 < x \leq 10$. What is going on? Explain.
86. Graph $y = \ln|\sin x|$ in the window $0 \leq x \leq 22$, $-2 \leq y \leq 0$. Explain what you see. How could you change the formula to turn the arches upside down?
87. a. Graph $y = \sin x$ and the curves $y = \ln(a + \sin x)$ for $a = 2, 4, 8, 20$, and 50 together for $0 \leq x \leq 23$.
b. Why do the curves flatten as a increases? (Hint: Find an a -dependent upper bound for $|y'|$.)
88. Does the graph of $y = \sqrt{x} - \ln x$, $x > 0$, have an inflection point? Try to answer the question (a) by graphing, (b) by using calculus.

7.3 The Exponential Function

Having developed the theory of the function $\ln x$, we introduce the exponential function $\exp x = e^x$ as the inverse of $\ln x$. We study its properties and compute its derivative and integral. Knowing its derivative, we prove the power rule to differentiate x^n when n is any real number, rational or irrational.

The Inverse of $\ln x$ and the Number e

The function $\ln x$, being an increasing function of x with domain $(0, \infty)$ and range $(-\infty, \infty)$, has an inverse $\ln^{-1} x$ with domain $(-\infty, \infty)$ and range $(0, \infty)$. The graph of $\ln^{-1} x$ is the graph of $\ln x$ reflected across the line $y = x$. As you can see in Figure 7.11,

$\lim_{x \rightarrow \infty} \ln^{-1} x = \infty$ and $\lim_{x \rightarrow -\infty} \ln^{-1} x = 0$.

The function $\ln^{-1} x$ is also denoted by $\exp x$.

In Section 7.2 we defined the number e by the equation $\ln(e) = 1$, so $e = \ln^{-1}(1) = \exp(1)$. Although e is not a rational number, later in this section we see one way to express it as a limit. In Chapter 11, we will calculate its value with a computer to as many places of accuracy as we want with a different formula (Section 11.9, Example 6). To 15 places,

$e = 2.718281828459045$.

The Function $y = e^x$

We can raise the number e to a rational power r in the usual way:

$e^2 = e \cdot e, \quad e^{-2} = \frac{1}{e^2}, \quad e^{1/2} = \sqrt{e},$

and so on. Since e is positive, e^r is positive too. Thus, e^r has a logarithm. When we take the logarithm, we find that

$\ln e^r = r \ln e = r \cdot 1 = r.$

Since $\ln x$ is one-to-one and $\ln(\ln^{-1} r) = r$, this equation tells us that

$e^r = \ln^{-1} r = \exp r \quad \text{for } r \text{ rational.} \tag{1}$

We have not yet found a way to give an obvious meaning to e^x for x irrational. But $\ln^{-1} x$ has meaning for any x , rational or irrational. So Equation (1) provides a way to extend the definition of e^x to irrational values of x . The function $\ln^{-1} x$ is defined for all x , so we use it to assign a value to e^x at every point where e^x had no previous definition.

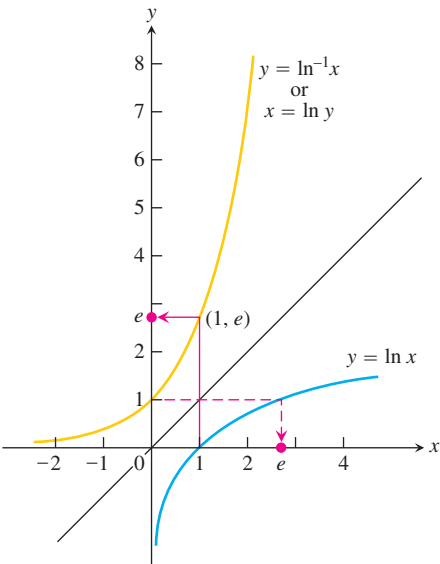


FIGURE 7.11 The graphs of $y = \ln x$ and $y = \ln^{-1} x = \exp x$. The number e is $\ln^{-1} 1 = \exp(1)$.

Typical values of e^x

x	e^x (rounded)
-1	0.37
0	1
1	2.72
2	7.39
10	22026
100	2.6881×10^{43}

DEFINITION The Natural Exponential Function

For every real number x , $e^x = \ln^{-1} x = \exp x$.

For the first time we have a precise meaning for an irrational exponent. Usually the exponential function is denoted by e^x rather than $\exp x$. Since $\ln x$ and e^x are inverses of one another, we have

Inverse Equations for e^x and $\ln x$

$$e^{\ln x} = x \quad (\text{all } x > 0) \quad (2)$$

$$\ln(e^x) = x \quad (\text{all } x) \quad (3)$$

Transcendental Numbers and Transcendental Functions

Numbers that are solutions of polynomial equations with rational coefficients are called **algebraic**: -2 is algebraic because it satisfies the equation $x + 2 = 0$, and $\sqrt{3}$ is algebraic because it satisfies the equation $x^2 - 3 = 0$. Numbers that are not algebraic are called **transcendental**, like e and π . In 1873, Charles Hermite proved the transcendence of e in the sense that we describe. In 1882, C.L.F. Lindemann proved the transcendence of π .

Today, we call a function $y = f(x)$ algebraic if it satisfies an equation of the form

$$P_n y^n + \cdots + P_1 y + P_0 = 0$$

in which the P 's are polynomials in x with rational coefficients. The function $y = 1/\sqrt{x+1}$ is algebraic because it satisfies the equation $(x+1)y^2 - 1 = 0$. Here the polynomials are $P_2 = x+1$, $P_1 = 0$, and $P_0 = -1$. Functions that are not algebraic are called transcendental.

The domain of $\ln x$ is $(0, \infty)$ and its range is $(-\infty, \infty)$. So the domain of e^x is $(-\infty, \infty)$ and its range is $(0, \infty)$.

EXAMPLE 1 Using the Inverse Equations

(a) $\ln e^2 = 2$

(b) $\ln e^{-1} = -1$

(c) $\ln \sqrt{e} = \frac{1}{2}$

(d) $\ln e^{\sin x} = \sin x$

(e) $e^{\ln 2} = 2$

(f) $e^{\ln(x^2+1)} = x^2 + 1$

(g) $e^{3 \ln 2} = e^{\ln 2^3} = e^{\ln 8} = 8$ One way

(h) $e^{3 \ln 2} = (e^{\ln 2})^3 = 2^3 = 8$ Another way ■

EXAMPLE 2 Solving for an Exponent

Find k if $e^{2k} = 10$.

Solution Take the natural logarithm of both sides:

$$e^{2k} = 10$$

$$\ln e^{2k} = \ln 10$$

$$2k = \ln 10 \quad \text{Eq. (3)}$$

$$k = \frac{1}{2} \ln 10. \quad \text{■}$$

The General Exponential Function a^x

Since $a = e^{\ln a}$ for any positive number a , we can think of a^x as $(e^{\ln a})^x = e^{x \ln a}$. We therefore make the following definition.

DEFINITION General Exponential Functions

For any numbers $a > 0$ and x , the exponential function with base a is

$$a^x = e^{x \ln a}.$$

When $a = e$, the definition gives $a^x = e^{x \ln a} = e^{x \ln e} = e^{x \cdot 1} = e^x$.

HISTORICAL BIOGRAPHY

Siméon Denis Poisson
(1781–1840)

EXAMPLE 3 Evaluating Exponential Functions

(a) $2^{\sqrt{3}} = e^{\sqrt{3} \ln 2} \approx e^{1.20} \approx 3.32$

(b) $2^{\pi} = e^{\pi \ln 2} \approx e^{2.18} \approx 8.8$ ■

We study the calculus of general exponential functions and their inverses in the next section. Here we need the definition in order to discuss the laws of exponents for e^x .

Laws of Exponents

Even though e^x is defined in a seemingly roundabout way as $\ln^{-1} x$, it obeys the familiar laws of exponents from algebra. Theorem 3 shows us that these laws are consequences of the definitions of $\ln x$ and e^x .

THEOREM 3 Laws of Exponents for e^x

For all numbers x , x_1 , and x_2 , the natural exponential e^x obeys the following laws:

1. $e^{x_1} \cdot e^{x_2} = e^{x_1+x_2}$
2. $e^{-x} = \frac{1}{e^x}$
3. $\frac{e^{x_1}}{e^{x_2}} = e^{x_1-x_2}$
4. $(e^{x_1})^{x_2} = e^{x_1 x_2} = (e^{x_2})^{x_1}$

Proof of Law 1 Let

$$y_1 = e^{x_1} \quad \text{and} \quad y_2 = e^{x_2}. \quad (4)$$

Then

$$\begin{aligned} x_1 &= \ln y_1 \quad \text{and} \quad x_2 = \ln y_2 && \text{Take logs of both sides of Eqs. (4).} \\ x_1 + x_2 &= \ln y_1 + \ln y_2 \\ &= \ln y_1 y_2 && \text{Product Rule for logarithms} \\ e^{x_1+x_2} &= e^{\ln y_1 y_2} && \text{Exponentiate.} \\ &= y_1 y_2 && e^{\ln u} = u \\ &= e^{x_1} e^{x_2}. \end{aligned}$$

■

The proof of Law 4 is similar. Laws 2 and 3 follow from Law 1 (Exercise 78).

EXAMPLE 4 Applying the Exponent Laws

(a) $e^{x+\ln 2} = e^x \cdot e^{\ln 2} = 2e^x$ Law 1

(b) $e^{-\ln x} = \frac{1}{e^{\ln x}} = \frac{1}{x}$ Law 2

(c) $\frac{e^{2x}}{e} = e^{2x-1}$ Law 3

(d) $(e^3)^x = e^{3x} = (e^x)^3$ Law 4 ■

Theorem 3 is also valid for a^x , the exponential function with base a . For example,

$$\begin{aligned}
 a^{x_1} \cdot a^{x_2} &= e^{x_1 \ln a} \cdot e^{x_2 \ln a} && \text{Definition of } a^x \\
 &= e^{x_1 \ln a + x_2 \ln a} && \text{Law 1} \\
 &= e^{(x_1 + x_2) \ln a} && \text{Factor } \ln a \\
 &= a^{x_1 + x_2}. && \text{Definition of } a^x
 \end{aligned}$$

The Derivative and Integral of e^x

The exponential function is differentiable because it is the inverse of a differentiable function whose derivative is never zero (Theorem 1). We calculate its derivative using Theorem 1 and our knowledge of the derivative of $\ln x$. Let

$$f(x) = \ln x \quad \text{and} \quad y = e^x = \ln^{-1} x = f^{-1}(x).$$

Then,

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx}(e^x) = \frac{d}{dx} \ln^{-1} x \\
 &= \frac{d}{dx} f^{-1}(x) \\
 &= \frac{1}{f'(f^{-1}(x))} && \text{Theorem 1} \\
 &= \frac{1}{f'(e^x)} && f^{-1}(x) = e^x \\
 &= \frac{1}{\left(\frac{1}{e^x}\right)} && f'(z) = \frac{1}{z} \text{ with } z = e^x \\
 &= e^x.
 \end{aligned}$$

That is, for $y = e^x$, we find that $dy/dx = e^x$ so the natural exponential function e^x is its own derivative. We will see in Section 7.5 that the only functions that behave this way are constant multiples of e^x . In summary,

$$\frac{d}{dx} e^x = e^x \quad (5)$$

EXAMPLE 5 Differentiating an Exponential

$$\begin{aligned}
 \frac{d}{dx}(5e^x) &= 5 \frac{d}{dx} e^x \\
 &= 5e^x
 \end{aligned}$$

The Chain Rule extends Equation (5) in the usual way to a more general form.

If u is any differentiable function of x , then

$$\frac{d}{dx} e^u = e^u \frac{du}{dx}. \quad (6)$$

EXAMPLE 6 Applying the Chain Rule with Exponentials

$$(a) \quad \frac{d}{dx} e^{-x} = e^{-x} \frac{d}{dx} (-x) = e^{-x}(-1) = -e^{-x} \quad \text{Eq. (6) with } u = -x$$

$$(b) \quad \frac{d}{dx} e^{\sin x} = e^{\sin x} \frac{d}{dx} (\sin x) = e^{\sin x} \cdot \cos x \quad \text{Eq. (6) with } u = \sin x \quad \blacksquare$$

The integral equivalent of Equation (6) is

$$\int e^u du = e^u + C.$$

EXAMPLE 7 Integrating Exponentials

$$(a) \quad \begin{aligned} \int_0^{\ln 2} e^{3x} dx &= \int_0^{\ln 8} e^u \cdot \frac{1}{3} du & u = 3x, \quad \frac{1}{3} du = dx, \quad u(0) = 0, \\ &= \frac{1}{3} \int_0^{\ln 8} e^u du & u(\ln 2) = 3 \ln 2 = \ln 2^3 = \ln 8 \\ &= \frac{1}{3} e^u \Big|_0^{\ln 8} \\ &= \frac{1}{3} (8 - 1) = \frac{7}{3} \end{aligned}$$

$$(b) \quad \begin{aligned} \int_0^{\pi/2} e^{\sin x} \cos x dx &= e^{\sin x} \Big|_0^{\pi/2} & \text{Antiderivative from Example 6} \\ &= e^1 - e^0 = e - 1 \end{aligned} \quad \blacksquare$$

EXAMPLE 8 Solving an Initial Value Problem

Solve the initial value problem

$$e^y \frac{dy}{dx} = 2x, \quad x > \sqrt{3}; \quad y(2) = 0.$$

Solution We integrate both sides of the differential equation with respect to x to obtain

$$e^y = x^2 + C.$$

We use the initial condition $y(2) = 0$ to determine C :

$$\begin{aligned} C &= e^0 - (2)^2 \\ &= 1 - 4 = -3. \end{aligned}$$

This completes the formula for e^y :

$$e^y = x^2 - 3.$$

To find y , we take logarithms of both sides:

$$\begin{aligned} \ln e^y &= \ln(x^2 - 3) \\ y &= \ln(x^2 - 3). \end{aligned}$$

Notice that the solution is valid for $x > \sqrt{3}$.

Let's check the solution in the original equation.

$$\begin{aligned} e^y \frac{dy}{dx} &= e^y \frac{d}{dx} \ln(x^2 - 3) && \text{Derivative of } \ln(x^2 - 3) \\ &= e^y \frac{2x}{x^2 - 3} && \\ &= e^{\ln(x^2 - 3)} \frac{2x}{x^2 - 3} && y = \ln(x^2 - 3) \\ &= (x^2 - 3) \frac{2x}{x^2 - 3} && e^{\ln y} = y \\ &= 2x. \end{aligned}$$

The solution checks. ■

The Number e Expressed as a Limit

We have defined the number e as the number for which $\ln e = 1$, or the value $\exp(1)$. We see that e is an important constant for the logarithmic and exponential functions, but what is its numerical value? The next theorem shows one way to calculate e as a limit.

THEOREM 4 The Number e as a Limit

The number e can be calculated as the limit

$$e = \lim_{x \rightarrow 0} (1 + x)^{1/x}.$$

Proof If $f(x) = \ln x$, then $f'(x) = 1/x$, so $f'(1) = 1$. But, by the definition of derivative,

$$\begin{aligned} f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{x \rightarrow 0} \frac{f(1+x) - f(1)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\ln(1+x) - \ln 1}{x} = \lim_{x \rightarrow 0} \frac{1}{x} \ln(1+x) && \ln 1 = 0 \\ &= \lim_{x \rightarrow 0} \ln(1+x)^{1/x} = \ln \left[\lim_{x \rightarrow 0} (1+x)^{1/x} \right] && \ln \text{ is continuous.} \end{aligned}$$

Because $f'(1) = 1$, we have

$$\ln \left[\lim_{x \rightarrow 0} (1 + x)^{1/x} \right] = 1$$

Therefore,

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e \quad \ln e = 1 \text{ and } \ln \text{ is one-to-one}$$

By substituting $y = 1/x$, we can also express the limit in Theorem 4 as

$$e = \lim_{y \rightarrow \infty} \left(1 + \frac{1}{y} \right)^y. \quad (7)$$

At the beginning of the section we noted that $e = 2.718281828459045$ to 15 decimal places.

The Power Rule (General Form)

We can now define x^n for any $x > 0$ and any real number n as $x^n = e^{n \ln x}$. Therefore, the n in the equation $\ln x^n = n \ln x$ no longer needs to be rational—it can be any number as long as $x > 0$:

$$\ln x^n = \ln (e^{n \ln x}) = n \ln x \quad \ln e^u = u, \text{ any } u$$

Together, the law $a^x/a^y = a^{x-y}$ and the definition $x^n = e^{n \ln x}$ enable us to establish the Power Rule for differentiation in its final form. Differentiating x^n with respect to x gives

$$\begin{aligned} \frac{d}{dx} x^n &= \frac{d}{dx} e^{n \ln x} && \text{Definition of } x^n, \ x > 0 \\ &= e^{n \ln x} \cdot \frac{d}{dx} (n \ln x) && \text{Chain Rule for } e^u \\ &= x^n \cdot \frac{n}{x} && \text{The definition again} \\ &= nx^{n-1}. \end{aligned}$$

In short, as long as $x > 0$,

$$\frac{d}{dx} x^n = nx^{n-1}.$$

The Chain Rule extends this equation to the Power Rule's general form.

Power Rule (General Form)

If u is a positive differentiable function of x and n is any real number, then u^n is a differentiable function of x and

$$\frac{d}{dx} u^n = nu^{n-1} \frac{du}{dx}.$$

EXAMPLE 9 Using the Power Rule with Irrational Powers

(a) $\frac{d}{dx}x^{\sqrt{2}} = \sqrt{2}x^{\sqrt{2}-1} \quad (x > 0)$

(b)
$$\begin{aligned}\frac{d}{dx}(2 + \sin 3x)^\pi &= \pi(2 + \sin 3x)^{\pi-1}(\cos 3x) \cdot 3 \\ &= 3\pi(2 + \sin 3x)^{\pi-1}(\cos 3x).\end{aligned}$$

■

EXERCISES 7.3

Algebraic Calculations with the Exponential and Logarithm

Find simpler expressions for the quantities in Exercises 1–4.

1. a. $e^{\ln 7.2}$ b. $e^{-\ln x^2}$ c. $e^{\ln x - \ln y}$
2. a. $e^{\ln(x^2 + y^2)}$ b. $e^{-\ln 0.3}$ c. $e^{\ln \pi x - \ln 2}$
3. a. $2 \ln \sqrt{e}$ b. $\ln(\ln e^e)$ c. $\ln(e^{-x^2 - y^2})$
4. a. $\ln(e^{\sec \theta})$ b. $\ln(e^{(e^x)})$ c. $\ln(e^{2 \ln x})$

Solving Equations with Logarithmic or Exponential Terms

In Exercises 5–10, solve for y in terms of t or x , as appropriate.

5. $\ln y = 2t + 4$ 6. $\ln y = -t + 5$
7. $\ln(y - 40) = 5t$ 8. $\ln(1 - 2y) = t$
9. $\ln(y - 1) - \ln 2 = x + \ln x$
10. $\ln(y^2 - 1) - \ln(y + 1) = \ln(\sin x)$

In Exercises 11 and 12, solve for k .

11. a. $e^{2k} = 4$ b. $100e^{10k} = 200$ c. $e^{k/1000} = a$
12. a. $e^{5k} = \frac{1}{4}$ b. $80e^k = 1$ c. $e^{(\ln 0.8)k} = 0.8$

In Exercises 13–16, solve for t .

13. a. $e^{-0.3t} = 27$ b. $e^{kt} = \frac{1}{2}$ c. $e^{(\ln 0.2)t} = 0.4$
14. a. $e^{-0.01t} = 1000$ b. $e^{kt} = \frac{1}{10}$ c. $e^{(\ln 2)t} = \frac{1}{2}$
15. $e^{\sqrt{t}} = x^2$ 16. $e^{(x^2)}e^{(2x+1)} = e^t$

Derivatives

In Exercises 17–36, find the derivative of y with respect to x , t , or θ , as appropriate.

17. $y = e^{-5x}$ 18. $y = e^{2x/3}$
19. $y = e^{5-7x}$ 20. $y = e^{(4\sqrt{x}+x^2)}$
21. $y = xe^x - e^x$ 22. $y = (1 + 2x)e^{-2x}$
23. $y = (x^2 - 2x + 2)e^x$ 24. $y = (9x^2 - 6x + 2)e^{3x}$
25. $y = e^\theta(\sin \theta + \cos \theta)$ 26. $y = \ln(3\theta e^{-\theta})$

27. $y = \cos(e^{-\theta^2})$

29. $y = \ln(3te^{-t})$

31. $y = \ln\left(\frac{e^\theta}{1 + e^\theta}\right)$

33. $y = e^{(\cos t + \ln t)}$

35. $y = \int_0^{\ln x} \sin e^t dt$

28. $y = \theta^3 e^{-2\theta} \cos 5\theta$

30. $y = \ln(2e^{-t} \sin t)$

32. $y = \ln\left(\frac{\sqrt{\theta}}{1 + \sqrt{\theta}}\right)$

34. $y = e^{\sin t}(\ln t^2 + 1)$

36. $y = \int_{e^{4\sqrt{x}}}^{e^{2x}} \ln t dt$

In Exercises 37–40, find dy/dx .

37. $\ln y = e^y \sin x$ 38. $\ln xy = e^{x+y}$
39. $e^{2x} = \sin(x + 3y)$ 40. $\tan y = e^x + \ln x$

Integrals

Evaluate the integrals in Exercises 41–62.

41. $\int (e^{3x} + 5e^{-x}) dx$

43. $\int_{\ln 2}^{\ln 3} e^x dx$

45. $\int 8e^{(x+1)} dx$

47. $\int_{\ln 4}^{\ln 9} e^{x/2} dx$

49. $\int \frac{e^{\sqrt{r}}}{\sqrt{r}} dr$

51. $\int 2t e^{-t^2} dt$

53. $\int \frac{e^{1/x}}{x^2} dx$

55. $\int_0^{\pi/4} (1 + e^{\tan \theta}) \sec^2 \theta d\theta$

57. $\int e^{\sec \pi t} \sec \pi t \tan \pi t dt$

58. $\int e^{\csc(\pi+t)} \csc(\pi+t) \cot(\pi+t) dt$

42. $\int (2e^x - 3e^{-2x}) dx$

44. $\int_{-\ln 2}^0 e^{-x} dx$

46. $\int 2e^{(2x-1)} dx$

48. $\int_0^{\ln 16} e^{x/4} dx$

50. $\int \frac{e^{-\sqrt{r}}}{\sqrt{r}} dr$

52. $\int t^3 e^{(t^4)} dt$

54. $\int \frac{e^{-1/x^2}}{x^3} dx$

56. $\int_{\pi/4}^{\pi/2} (1 + e^{\cot \theta}) \csc^2 \theta d\theta$

$$59. \int_{\ln(\pi/6)}^{\ln(\pi/2)} 2e^v \cos e^v dv \quad 60. \int_0^{\sqrt{\ln \pi}} 2x e^{x^2} \cos(e^{x^2}) dx$$

$$61. \int \frac{e^r}{1+e^r} dr \quad 62. \int \frac{dx}{1+e^x}$$

Initial Value Problems

Solve the initial value problems in Exercises 63–66.

$$63. \frac{dy}{dt} = e^t \sin(e^t - 2), \quad y(\ln 2) = 0$$

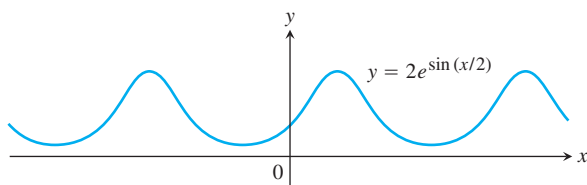
$$64. \frac{dy}{dt} = e^{-t} \sec^2(\pi e^{-t}), \quad y(\ln 4) = 2/\pi$$

$$65. \frac{d^2y}{dx^2} = 2e^{-x}, \quad y(0) = 1 \quad \text{and} \quad y'(0) = 0$$

$$66. \frac{d^2y}{dt^2} = 1 - e^{2t}, \quad y(1) = -1 \quad \text{and} \quad y'(1) = 0$$

Theory and Applications

67. Find the absolute maximum and minimum values of $f(x) = e^x - 2x$ on $[0, 1]$.
68. Where does the periodic function $f(x) = 2e^{\sin(x/2)}$ take on its extreme values and what are these values?



69. Find the absolute maximum value of $f(x) = x^2 \ln(1/x)$ and say where it is assumed.

- T** 70. Graph $f(x) = (x - 3)^2 e^x$ and its first derivative together. Comment on the behavior of f in relation to the signs and values of f' . Identify significant points on the graphs with calculus, as necessary.

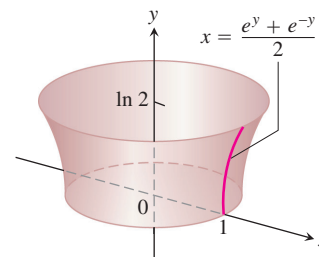
71. Find the area of the “triangular” region in the first quadrant that is bounded above by the curve $y = e^{2x}$, below by the curve $y = e^x$, and on the right by the line $x = \ln 3$.

72. Find the area of the “triangular” region in the first quadrant that is bounded above by the curve $y = e^{x/2}$, below by the curve $y = e^{-x/2}$, and on the right by the line $x = 2 \ln 2$.

73. Find a curve through the origin in the xy -plane whose length from $x = 0$ to $x = 1$ is

$$L = \int_0^1 \sqrt{1 + \frac{1}{4} e^x} dx.$$

74. Find the area of the surface generated by revolving the curve $x = (e^y + e^{-y})/2$, $0 \leq y \leq \ln 2$, about the y -axis.



75. a. Show that $\int \ln x dx = x \ln x - x + C$.

- b. Find the average value of $\ln x$ over $[1, e]$.

76. Find the average value of $f(x) = 1/x$ on $[1, 2]$.

77. **The linearization of e^x at $x = 0$**

- a. Derive the linear approximation $e^x \approx 1 + x$ at $x = 0$.

- T** b. Estimate to five decimal places the magnitude of the error involved in replacing e^x by $1 + x$ on the interval $[0, 0.2]$.

- T** c. Graph e^x and $1 + x$ together for $-2 \leq x \leq 2$. Use different colors, if available. On what intervals does the approximation appear to overestimate e^x ? Underestimate e^x ?

78. **Laws of Exponents**

- a. Starting with the equation $e^{x_1} e^{x_2} = e^{x_1+x_2}$, derived in the text, show that $e^{-x} = 1/e^x$ for any real number x . Then show that $e^{x_1}/e^{x_2} = e^{x_1-x_2}$ for any numbers x_1 and x_2 .

- b. Show that $(e^{x_1})^{x_2} = e^{x_1 x_2} = (e^{x_2})^{x_1}$ for any numbers x_1 and x_2 .

- T** 79. **A decimal representation of e** Find e to as many decimal places as your calculator allows by solving the equation $\ln x = 1$.

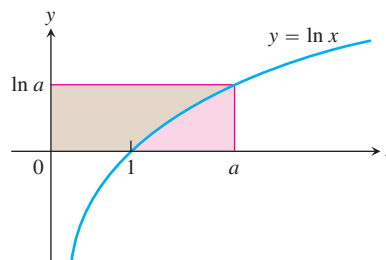
- T** 80. **The inverse relation between e^x and $\ln x$** Find out how good your calculator is at evaluating the composites

$$e^{\ln x} \quad \text{and} \quad \ln(e^x).$$

81. Show that for any number $a > 1$

$$\int_1^a \ln x dx + \int_0^{\ln a} e^y dy = a \ln a.$$

(See accompanying figure.)



82. **The geometric, logarithmic, and arithmetic mean inequality**

- a. Show that the graph of e^x is concave up over every interval of x -values.